

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The global proliferation of digital technologies over the past two decades has fundamentally transformed the manner in which commercial and cultural events are organized, ticketed, and managed. In developed economies, the transition from paper-based to digital ticketing systems has become largely complete, with platforms such as Eventbrite, Ticketmaster, and StubHub processing hundreds of millions of ticket transactions annually. These systems have demonstrated measurable benefits across the entire event value chain: organizers gain real-time revenue visibility and attendance analytics, attendees gain convenience and security, and gate operations become faster and more fraud-resistant. The underlying technology — principally QR code validation, mobile payment integration, and cloud-based event management — is now mature, well-documented, and available as open-source building blocks that any competent developer team can deploy.

Sub-Saharan Africa, however, presents a different and considerably more complex landscape for digital service deployment. While mobile phone penetration has grown dramatically across the continent — reaching over 46% of the population according to the GSMA Mobile Economy Report 2024 — the infrastructure supporting these devices remains uneven and often inadequate for conventional web application architectures. Electricity supply is intermittent in many regions. Mobile internet speeds are low and bandwidth is expensive relative to local incomes. Banking penetration remains limited, but mobile money services have emerged as a genuine and widespread alternative payment channel, particularly in East and Central Africa. Against this backdrop, the event management sector across much of the continent continues to rely on paper ticket systems that have changed little in decades.

South Sudan, the world's youngest nation, presents these challenges in particularly acute form. According to the International Telecommunication Union ICT Development Index 2023, South Sudan ranks among the lowest-connected countries in the world, with fewer than ten percent of the population having regular internet access. Mobile data connectivity is concentrated in Juba, the capital, and progressively less reliable in

secondary cities such as Wau, Malakal, and Yei. Fixed-line broadband infrastructure is negligible. Electricity supply, even in Juba, depends heavily on diesel generators that are expensive to operate and vulnerable to fuel supply disruptions. Despite this infrastructure deficit, South Sudan has a vibrant and growing events ecosystem. University of Juba graduation ceremonies annually draw thousands of attendees. South Sudan Premier League football matches at Juba Stadium attract passionate crowds. Church conferences organized by major evangelical and Catholic parishes gather thousands of participants from across the country. Music concerts in Juba, while still emerging as a commercial sector, have grown significantly in the post-conflict stabilization period.

All of these events share a common operational architecture: paper tickets, informal cash transactions at vendor points scattered across the city, and human visual inspection at entry gates. This architecture is not merely inefficient — it is systematically exploitable. Paper tickets can be photocopied using any basic colour printer available at the numerous small printing shops found throughout Juba. Visual verification under the pressure and noise of a busy event gate cannot reliably detect high-quality duplicates. There is no mechanism to prevent a valid-looking copy from being admitted; there is no audit trail of who entered; and there is no data that allows the organizer to understand what actually happened at their event. According to reports gathered from Juba concert promoters during the contextual research phase of this project, fraudulent entry rates at major events range from ten to thirty percent of total attendance — a substantial and preventable revenue loss that directly undermines the economic viability of event organization in South Sudan.

The emergence and growth of mobile money services in South Sudan provides a critical enabling condition that did not exist even five years ago. MTN Mobile Money and Airtel Money both operate in South Sudan, offering USSD-based transaction capabilities that function on basic handsets without internet connectivity. Mobile money penetration, while lower than in markets such as Kenya or Uganda, is growing particularly among urban and peri-urban populations — precisely the demographics most likely to attend commercial events. This provides the payment infrastructure upon which a digital ticketing system can be built without requiring bank cards, online payment gateways, or the levels of internet connectivity that conventional e-commerce platforms demand.

The case for a QR code-based digital ticketing system for South Sudan is therefore grounded not in technological novelty but in operational necessity. QR codes represent an

ideal validation technology for this context: they are generated at zero cost using standard open-source libraries, they can be validated by any smartphone camera without requiring dedicated hardware, they can be cryptographically secured against forgery through well-established token-signing algorithms, and they can be validated offline through local data caching mechanisms that function independently of network connectivity at the event gate. The combination of QR-based validation, mobile money payment integration, SMS ticket delivery fallback, and offline-capable gate scanning addresses each of the core failure modes of the existing paper ticket system in ways that are technically achievable, economically viable, and operationally practical within the South Sudan context.

This project documents the complete software engineering lifecycle of such a system: from contextual analysis and requirements elicitation through system design, implementation, security analysis, testing, and evaluation. The resulting artefact — a functioning event booking and ticketing system with QR code validation — represents a practical contribution to the digitalization of South Sudan's event management sector and a demonstrable application of software engineering principles to the challenges of resource-constrained African deployment environments.

1.2 Problem Statement

South Sudan's event management sector operates without a reliable, secure, or analytically capable ticketing infrastructure. The existing system, based entirely on paper tickets sold through informal vendor networks and verified through manual gate inspection, exhibits a cluster of interrelated failures that collectively represent a significant and solvable problem for event organizers, attendees, and the broader development of the events economy in the country.

The most economically damaging manifestation of the current system's failure is the systematic leakage of ticket revenue through counterfeiting and unauthorized duplication. Paper tickets in South Sudan are produced without cryptographic security features, without unique machine-readable identifiers, and without any mechanism for detecting whether a ticket has already been used for entry. Any ticket that can be visually observed can be photographically reproduced using equipment that is widely available and inexpensive. Promoters at Juba events report that between ten and thirty percent of apparent attendees may have entered using fraudulent documents, representing direct revenue losses that accumulate across every event in the calendar. For an organizer

running a 2,000-seat concert at 50 SSP per ticket, a twenty percent fraud rate represents a loss of 20,000 SSP per event — revenue that could have funded production costs, artist fees, or venue improvements.

Manual gate verification is not only vulnerable to fraud — it is fundamentally unable to scale efficiently to large event volumes. Human visual inspection of tickets is slow, particularly when the ticket design does not include easily machine-verifiable elements. At high-demand events such as South Sudan Premier League finals or large church conference sessions, queue buildup at entry gates creates crowd management challenges that have documented safety implications. Gate congestion contributes directly to pushing and jostling behavior as large numbers of attendees attempt to enter within a short pre-event window. In the absence of any capacity control mechanism linked to ticket sales, there is no upper bound on the number of people who can attempt to enter a venue simultaneously.

The current system also systematically excludes a significant potential market segment: attendees who are not physically located in Juba or near authorized ticket vendors. An individual in Wau who wishes to attend a concert in Juba must either travel to the capital before the event to purchase a ticket in person, or rely on informal intermediaries whose reliability cannot be guaranteed. There is no mechanism for remote ticket purchase, digital delivery, or safe transfer of payment to the organizer from a distance. This geographic barrier limits event revenue, restricts attendance to those with physical proximity to vendor networks, and prevents the development of a national events market that could generate economic activity across multiple cities simultaneously.

Beyond fraud, exclusion, and congestion, the current system generates no data. An organizer completing a successful event cannot answer basic questions: exactly how many people attended? At what time was peak entry? How many tickets were sold versus used versus fraudulently duplicated? What proportion of attendees came from which sources? This absence of analytics prevents effective planning, makes it impossible to demonstrate attendance figures to sponsors or venues, and creates a black box around an activity that, in other markets, would generate commercially valuable insights. The absence of data is not merely an inconvenience — it is a structural impediment to the growth and professionalization of South Sudan's events industry.

In summary, the problem that this project addresses is the complete absence of a digital, secure, data-generating, mobile money-integrated, and offline-capable event ticketing system designed for the specific infrastructure constraints of South Sudan. This problem is not addressed by importing solutions designed for Western markets — the payment infrastructure, connectivity conditions, device demographics, and user literacy profiles of South Sudan require a purpose-built solution grounded in local context. The proposed Event Booking and Ticketing System with QR Code Validation addresses each dimension of this problem through specific design choices documented in the subsequent chapters of this report.

1.3 Objectives

1.3.1 General Objective

The general objective of this project is to design, develop, and evaluate a web-based Event Booking and Ticketing System with QR Code Validation that addresses the documented failures of paper-based event management in South Sudan, providing event organizers with a secure, data-generating, and operationally practical platform for managing ticket sales, payment processing, and entry validation in the specific infrastructure environment of South Sudan.

1.3.2 Specific Objectives

- To analyse the current state of event management and ticketing in South Sudan, identifying the specific operational failures, security vulnerabilities, and user needs that a digital system must address.
- To design a system architecture that supports online ticket booking with mobile money payment integration through MTN Mobile Money and Airtel Money, accommodating the payment channel realities of South Sudanese event attendees.
- To implement a QR code generation module that produces cryptographically signed, unique, and tamper-resistant ticket tokens using HMAC-SHA256 JSON Web Token standards, ensuring that each generated ticket is both authentic and unforgeable.
- To develop an offline-capable QR validation module for gate agents that functions without internet connectivity through an IndexedDB-based local cache, synchronizing scan records with the central server when connectivity is restored.

- To implement a comprehensive attendance tracking and analytics module that provides event organizers with real-time entry statistics, peak attendance data, and post-event reporting capabilities.
- To integrate an SMS ticket delivery system using the Africa's Talking API that delivers QR tickets as accessible links to feature phone users without email addresses or smartphone applications.
- To evaluate the system's performance against measurable thresholds including QR validation accuracy, scan response time, booking completion rate, fraud detection effectiveness, and user satisfaction.

1.4 Research Questions

This study is guided by the following research questions, which collectively frame the technical, contextual, and evaluative dimensions of the project:

- What are the specific operational failures, security vulnerabilities, and user experience limitations of the current paper-based event ticketing system in South Sudan, and how do these translate into measurable losses for organizers, attendees, and the broader events sector?
- How can a cryptographically secured QR code validation system be designed and implemented to prevent ticket fraud, duplicate entry, and unauthorized access in the specific device and connectivity context of South Sudanese event venues?
- To what extent can mobile money payment integration with MTN Mobile Money and Airtel Money enable effective digital ticket sales among South Sudanese event attendees who lack conventional banking services or online payment capabilities?
- How can an offline-first architecture be implemented to ensure that gate-side QR validation functions reliably without internet connectivity, and what are the limitations of such an architecture in the specific context of multi-gate, high-volume South Sudan events?
- What performance metrics — including QR scan accuracy, response time, booking completion rate, and fraud detection success rate — does the implemented system achieve, and how do these compare to the documented performance of existing ticketing systems in comparable African markets?

1.5 Scope of the Study

1.5.1 Geographical Scope

The geographical scope of this study is the Republic of South Sudan, with particular focus on event management activities occurring in Juba, the national capital. The contextual analysis, user research, and system design are grounded in the specific infrastructure conditions, user demographics, payment ecosystems, and event culture of South Sudan. While the system's technical architecture is applicable to other countries in the East and Central African region facing similar infrastructure constraints, the design decisions documented in this report — including the selection of mobile money providers, SMS gateway, connectivity assumptions, and user literacy considerations — are specifically calibrated for the South Sudan deployment context.

1.5.2 Content Scope

The content scope of this project encompasses the complete software engineering lifecycle of an event booking and ticketing system, from requirements analysis through system design, implementation, security analysis, testing, and performance evaluation. Specifically, the system covers: event creation and configuration by organizers; ticket category definition and capacity management; attendee registration and account management; ticket booking and mobile money payment processing; QR code generation and cryptographic token signing; multi-channel ticket delivery including in-app access and SMS fallback; gate-side QR scanning and validation with offline capability; attendance tracking and real-time analytics; fraud detection and duplicate scan prevention; and post-event reporting. The system does not cover live event streaming, social networking features, international multi-currency payments beyond South Sudanese mobile money providers, or integration with national identification systems.

1.6 Significance of the Study

The significance of this project spans multiple dimensions — technical, economic, operational, and social — and extends beyond the immediate academic context of a final year project to encompass genuine contributions to the digitalization of South Sudan's events sector and the broader body of knowledge on context-appropriate software engineering for resource-constrained African environments.

From the perspective of event organizers, the system provides a direct and measurable solution to revenue leakage through ticket fraud, which current evidence suggests costs the sector between ten and thirty percent of potential revenues per event. By eliminating paper tickets and replacing manual visual inspection with cryptographic QR validation, the system eliminates the most exploitable vulnerability in the current workflow. The analytics module provides organizers with attendance data that is currently entirely absent, enabling data-driven decisions about event planning, marketing, and capacity management that were previously impossible. For promoters managing concerts at Juba venues, church administrators organizing national conferences, and university officials planning graduation ceremonies, the practical operational benefits of the system are immediate and substantial.

For attendees, the significance lies primarily in accessibility and experience. The integration of mobile money payment removes the geographic barrier that currently excludes remote attendees from purchasing tickets without physical presence in Juba. The SMS ticket delivery fallback ensures that the system's benefits extend to feature phone users with limited digital literacy, not merely to the smartphone-owning urban minority. The faster gate processing that QR scanning enables relative to manual ticket inspection reduces queue times and the discomfort and safety risks associated with crowd buildup at entry points.

For the academic community in South Sudan and the broader African region, the project contributes a documented case study of offline-first architecture implementation using IndexedDB and service workers for a security-critical validation use case. The security architecture combining HMAC-SHA256 signed JWT tokens with server-side duplicate prevention demonstrates a practical approach to cryptographic ticket security that is achievable within undergraduate project scope and applicable to numerous related domains. These contributions add to the still-limited body of software engineering literature specifically addressing African deployment contexts.

For government and development organizations working on South Sudan's digital transformation agenda, the project demonstrates the technical and economic feasibility of deploying digital public services using existing mobile infrastructure — specifically mobile money and SMS — without requiring significant investment in fixed broadband or banking infrastructure. The feasibility analysis documented in Chapter Three provides

a replicable framework for assessing the viability of similar digital service deployments in comparable contexts.

1.7 Justification of the Study

The justification for undertaking this project derives from three converging sources of evidence: documented market failure in the South Sudan events sector, technological readiness demonstrated by the maturity of the relevant open-source tools and APIs, and the absence of existing solutions appropriately designed for the South Sudan context.

The market failure evidence is substantial and directly observable. The paper ticket system in use across South Sudan's events sector fails systematically and predictably on the dimensions of security, inclusivity, scalability, and data generation. These failures are not the result of temporary or contingent circumstances — they are structural characteristics of a technology that has not been updated to reflect the capabilities now available at negligible cost through mobile devices and open-source software. The organizer who loses twenty percent of concert revenue to counterfeit tickets, the attendee in Wau who cannot book a ticket without traveling to Juba, and the graduation administrator who cannot produce an accurate attendance figure for the university's records are all experiencing the consequences of a market that has not yet adapted to available technology.

The technological readiness argument is equally compelling. Every technical component required to build the proposed system is mature, well-documented, and available as open-source software. Node.js, PostgreSQL, React.js, JWT token libraries, QR generation and scanning libraries, and IndexedDB-based offline storage mechanisms are all actively maintained and have been deployed in production systems globally. The Africa's Talking SMS API is operational in South Sudan. MTN Mobile Money and Airtel Money both provide developer APIs for payment integration. There are no fundamental technical barriers to building the proposed system — the question is not whether it can be built, but whether it will be built with sufficient sensitivity to the South Sudan context to be actually usable in the conditions where it must operate.

The absence of contextually appropriate existing solutions completes the justification. International platforms such as Eventbrite and Ticketmaster are technically sophisticated but practically inaccessible to the South Sudan market: they require credit card or PayPal payment methods not available to most South Sudanese users, they depend on stable

internet connectivity at the point of booking and validation, and they are not designed to accommodate the specific cultural patterns — communal ticket purchasing, high walk-in rates, SMS-reliant demographics — of South Sudanese event culture. Local alternatives, where they exist, are informal, undocumented, and not built on secure engineering foundations. The research gap identified in Chapter Two confirms that no published work documents a ticketing system purpose-built for the South Sudan context with the full feature set proposed here.

1.8 Limitations of the Study

Every software engineering project operates within constraints that bound the scope of what can be achieved within the available time, resources, and technical capabilities. Acknowledging these limitations with intellectual honesty is an essential part of rigorous academic and professional practice, and reflects the design principle stated throughout this project: good engineering requires explicit attention to failure modes, not avoidance of them.

The most operationally significant limitation is the multi-gate offline duplicate admission risk inherent in any offline-first validation architecture. If two gate devices are simultaneously operating in offline mode — that is, without any network connectivity — and both devices scan the same QR code at different entry points before either device can synchronize its scan records with the central server, both instances of that QR code will be admitted. This is not a failure of implementation but a fundamental constraint of distributed systems operating without real-time consensus. The system mitigates this risk through gate-level ticket tier partitioning — assigning VIP tickets exclusively to one gate and general admission tickets to another, so that the same ticket is extremely unlikely to be presented at two gates simultaneously — but cannot eliminate it entirely without real-time network dependency, which would undermine the core offline functionality.

The SMS ticket delivery system, while functional and tested against the Africa's Talking API, is subject to delivery delays that can be significant in areas of South Sudan with poor mobile network coverage. An attendee who purchases a ticket shortly before an event and resides in an area with intermittent network reception may experience delays in receiving their SMS ticket link that could affect their ability to present the ticket at the gate. The system's mitigation — providing in-app ticket access that is independent of

SMS delivery — is effective for smartphone users but not for feature phone users who depend entirely on the SMS link.

The offline validation functionality, while implemented and tested for Android Progressive Web Applications, exhibits limitations on iOS devices due to Apple's historically restrictive handling of IndexedDB storage quotas in mobile Safari. Given that Android represents over ninety percent of smartphone usage in South Sudan, this is an acceptable limitation for the current project scope, but represents a known gap that would need to be addressed through a native iOS application in future development phases.

Finally, the evaluation conducted as part of this project is necessarily limited by the academic context. Performance testing was conducted in a controlled laboratory environment simulating concurrent user loads rather than under live event conditions with real attendees. Usability testing involved a sample of participants that, while selected to represent relevant user demographics, is smaller than would be ideal for a fully validated user experience study. These methodological constraints are inherent to undergraduate project research and do not invalidate the findings, but they do set appropriate boundaries on the generalizability of the performance metrics reported in Chapter Five.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The literature review presented in this chapter provides the theoretical, conceptual, and empirical foundations upon which the design and development decisions of the proposed system are grounded. A systematic review of academic literature, technical documentation, and case study evidence serves multiple purposes within this research: it establishes the theoretical frameworks that explain user adoption of digital ticketing systems in developing contexts, it surveys the conceptual domain of QR code technology, mobile payment systems, and fraud prevention architectures, it provides empirical comparative evidence from existing ticketing systems deployed in developed and African markets, and it identifies the specific research gaps that justify the development of a new system purpose-built for South Sudan. The review draws on peer-reviewed journal articles, conference proceedings, technical reports, and industry publications spanning the period 2015 to 2024, supplemented by technical documentation from standards bodies and API providers relevant to the proposed implementation.

2.2 Theoretical Review

2.2.1 Information Systems Theory

Information Systems Theory, as developed through the foundational contributions of Laudon and Laudon (2020) and subsequently elaborated in the context of developing nations by Heeks (2018), provides the broad conceptual framework within which the proposed ticketing system sits. The theory holds that information systems are not neutral technological artefacts but sociotechnical systems whose success or failure is determined by the fit between the system's design, the organizational processes it seeks to support, and the social and environmental context in which it is deployed. Heeks's influential ITPOSMO framework for ICT4D (Information and Communication Technology for Development) failure analysis identifies gaps between the design assumptions built into a system and the realities of the deployment context — including gaps in information content, technology assumptions, processes, objectives, staffing and skills, management structures, and other resources — as the primary predictors of system failure in developing country deployments.

This theoretical lens is directly relevant to the problem identified in Chapter One. International ticketing systems such as Eventbrite and Ticketmaster were designed with implicit assumptions about the deployment context — reliable internet connectivity, credit card payment infrastructure, English-literate users, smartphone device penetration — that do not hold in South Sudan. When these systems are evaluated against the ITPOSMO framework in the South Sudan context, multiple critical gaps become immediately apparent: the payment infrastructure gap (credit cards versus mobile money), the connectivity gap (reliable broadband versus intermittent 3G), the digital literacy gap (assumed app sophistication versus feature phone familiarity), and the hardware assumption gap (desktop-first design versus mobile-first reality). These gaps predict, with high confidence, the failure of direct system transplantation and justify the context-specific design approach taken in this project.

2.2.2 Technology Acceptance Model (TAM)

The Technology Acceptance Model, originally formulated by Davis (1989) and subsequently refined through numerous extensions and applications in African ICT contexts, provides a framework for understanding the conditions under which potential users will adopt and continue using a new technological system. TAM posits that adoption is primarily determined by two perceptual variables: perceived usefulness — the degree to which a person believes using the system will enhance their performance — and perceived ease of use — the degree to which they believe the system is free of effort. In its extensions relevant to developing country contexts, variables including social influence, facilitating conditions, and trust have been added as significant predictors of adoption behavior.

In the South Sudan context, TAM predicts several design imperatives for the proposed system. Perceived usefulness for organizers will be high if the system demonstrably reduces fraud losses and provides attendance data — both of which are measurable and communicable benefits. Perceived usefulness for attendees will depend on whether the system genuinely removes the effort of physical ticket purchase without introducing comparable effort in the form of technology complexity. Perceived ease of use will be critically impacted by the interface design choices made for the booking flow and the gate scanning application: interfaces calibrated for technically literate users will score poorly on ease of use for the demographically diverse South Sudanese attendee population. The

TAM framework thus directly informs the inclusive design principles applied in the frontend development phase of this project, particularly the decision to support SMS ticket delivery for feature phone users and to implement a gate interface designed for operators with limited digital training.

2.2.3 Digital Transformation Theory

Digital transformation theory, as articulated by Westerman, Bonnet, and McAfee (2014) in the context of business digitalization and subsequently applied to public sector and development contexts by Nambisan et al. (2019), describes the process by which digital technologies are used to create new or modify existing processes, culture, and customer experiences to meet changing business and market requirements. In the context of this project, digital transformation theory provides the macro-level justification for the transition from paper to digital ticketing in South Sudan: it is not merely a substitution of one tool for another, but a transformation of the entire value chain of event management — from isolated, cash-based, undocumented transactions to an integrated, data-generating, secure, and inclusive digital ecosystem.

Critically, digital transformation theory also cautions against transformation for transformation's sake. The literature consistently finds that successful digital transformations in developing contexts require careful attention to the digital readiness of the target population, the sustainability of the technical infrastructure adopted, and the alignment between the transformed process and the social context in which it operates. These cautions are directly reflected in the design principles of the proposed system: the retention of cash payment support through a Point-of-Sale mode for walk-in customers, the SMS fallback for feature phone users, and the offline-first architecture for gate validation all represent conscious decisions to transform the process incrementally rather than imposing a fully digital dependency that the infrastructure cannot reliably support.

2.3 Conceptual Review

2.3.1 QR Code Systems in Event Management

QR (Quick Response) codes, first developed by Denso Wave in 1994 for automotive parts tracking, have become the dominant technology for event ticket validation globally over the past decade. The academic literature on QR code applications in ticketing is well-established, with Erkan and Evans (2016) documenting the adoption trajectory in

European concert venues and Hsu and Lin (2018) providing a comprehensive analysis of QR ticket fraud prevention architectures. The core technical literature, including the ISO/IEC 18004 standard that defines the QR code specification, establishes the properties that make QR codes suitable for ticketing applications: high data density (up to 4,296 alphanumeric characters), four levels of error correction providing resilience against physical damage, omnidirectional readability eliminating the need for precise scanner alignment, and compatibility with standard smartphone camera hardware eliminating the need for dedicated scanning equipment.

However, the raw properties of QR codes as a data encoding medium do not, by themselves, provide ticket security. A QR code containing a plaintext ticket identifier provides no more security against counterfeiting than a printed serial number — any scanner application can read the identifier and any printer can reproduce the QR image. Security derives not from the QR code format itself but from the data encoded within it. The literature on cryptographic ticket security (Bhatt et al., 2017; Kumar and Singh, 2020) consistently identifies signed token architectures — specifically implementations based on the JSON Web Token standard — as the most appropriate approach for event ticket security in systems where offline validation is required. The proposed system's QR security architecture, described in detail in Chapter Four, directly implements the HMAC-SHA256 signed JWT approach recommended in this literature.

2.3.2 Online Ticketing Systems

Online ticketing systems represent a well-studied domain in information systems research, with literature spanning user experience design, payment system integration, fraud prevention, and scalability architecture. The general architecture of contemporary online ticketing systems follows a client-server model in which a web or mobile frontend communicates with a backend API that manages event data, ticket inventory, payment processing, and validation logic. Huang and Lin (2019) provide a comprehensive architectural analysis of high-availability ticketing system design, addressing the specific challenge of simultaneous ticket requests during high-demand event launches — a problem that requires atomic database transaction design to prevent overselling.

The literature on ticketing system localization for African markets is considerably thinner. Whereas the architecture of an online ticketing system is relatively universal, the specific requirements for payment integration, connectivity tolerance, device compatibility, and

user interface design vary substantially between markets. The paucity of literature specifically addressing event ticketing system design for East and Central African markets represents a research gap that this project addresses and contributes to.

2.3.3 Mobile Money Integration in Digital Commerce

Mobile money services have transformed digital commerce in sub-Saharan Africa, providing a payment infrastructure layer accessible to populations excluded from conventional banking systems. The academic literature on mobile money integration in digital services is extensive and growing, with particularly strong contributions from the Kenyan M-Pesa ecosystem which has generated a large body of empirical research on adoption, impact, and technical implementation. Osei-Assibey (2015) documented the impact of mobile money on financial inclusion in Ghana, demonstrating the degree to which USSD-based payment channels extend digital commerce access to lower-income and rural demographics. Jack and Suri (2014) provided landmark evidence from Kenya demonstrating the consumption-smoothing effects of mobile money access and the degree to which it enables participation in economic activities previously accessible only to the banked population.

Technical literature on mobile money API integration, including documentation from Africa's Talking (2024) and the M-Pesa Developer Platform (2023), describes the webhook-based payment confirmation architecture that the proposed system implements: a user-initiated payment request is processed by the mobile money provider, who then sends a callback (webhook) to the ticketing system server upon payment confirmation. This asynchronous architecture requires careful implementation of idempotency controls — ensuring that a duplicate callback for the same transaction is processed only once — and status polling mechanisms for transactions whose callbacks are delayed or lost. These implementation considerations are addressed in the system design documentation in Chapter Four.

2.3.4 Offline-First Application Architecture

Offline-first architecture — the design principle of building applications that function fully without network connectivity as the primary mode of operation, rather than as a degraded fallback — emerged as a response to the connectivity realities of mobile internet usage, particularly in regions where network coverage is intermittent or unreliable. The foundational technical literature on offline-first design, including the

Google Developers Progressive Web App documentation (2023) and Hood's influential treatment of the subject (2015), identifies IndexedDB as the appropriate client-side storage technology for offline data persistence in web applications, and service workers as the mechanism for intercepting network requests and serving cached responses during offline periods.

In the specific context of gate-side QR validation, the offline-first requirement creates a distinctive architectural challenge: the validation application must have access to the complete set of valid tickets for the current event stored locally on the gate device before the event begins, must be able to validate scanned tickets against this local dataset without any server communication, and must synchronize the scan records generated during offline operation back to the central server when connectivity is restored. The literature on this specific architecture — often termed 'pre-fetch and sync' — is less well developed than general offline-first design literature, representing an area where the implementation choices documented in this project make a genuine technical contribution.

2.4 Empirical Review

2.4.1 Eventbrite and Global Ticketing Platforms

Eventbrite, the leading global self-service event ticketing platform, provides the most extensively documented reference point for online ticketing system performance. Its annual reports and developer documentation describe a system handling over 300 million ticket transactions annually across 170 countries, with QR validation accuracy exceeding 99.5% and average scan response times below 0.5 seconds under optimal network conditions. Eventbrite's architecture, extensively documented in technical publications including an analysis by Kumar et al. (2019), employs a microservices model with distributed caching, horizontal auto-scaling, and real-time WebSocket updates for organizer dashboards. These performance benchmarks provide useful reference points for evaluating the proposed South Sudan system, though direct comparison must account for the substantially different operating conditions.

From the perspective of applicability to the South Sudan market, Eventbrite's critical limitation is its payment infrastructure: the platform requires credit cards, PayPal, or Stripe payment methods that are inaccessible to the overwhelming majority of South Sudanese event attendees. Eventbrite has made limited progress in integrating African

mobile money providers, and its interface design reflects the digital literacy assumptions of developed market users. These limitations are not failures of Eventbrite as a platform — they reflect the market it was designed to serve — but they confirm that direct deployment of Eventbrite in the South Sudan context would fail on the payment and usability dimensions identified by the ITPOSMO analysis.

2.4.2 Ticketmaster and Enterprise-Scale Systems

Ticketmaster represents the enterprise tier of event ticketing, focused on large-scale stadium and arena events with highly sophisticated fraud prevention architectures. Its Verified Fan programme and dynamic barcode rotation technology — which changes the visual representation of the ticket QR code every few seconds to prevent screenshot sharing — represents the state of the art in consumer ticket fraud prevention. Academic analysis of Ticketmaster's architecture (Chen and Wang, 2021) describes a system with essentially unlimited horizontal scalability, sophisticated bot prevention, and secondary market integration that collectively eliminate the operational vulnerabilities exploited in the South Sudan context. However, Ticketmaster's enterprise pricing model, credit card payment dependency, and complete absence of offline validation capability make it entirely inapplicable to the South Sudan deployment scenario.

2.4.3 African Ticketing Platforms: iTickets, TicketSasa, and Mtickets

Several African-specific ticketing platforms have emerged over the past decade, addressing some but not all of the limitations of global platforms in the African context. iTickets (South Africa) provides mobile money integration for South African payment methods and has demonstrated successful deployment for major music events in Johannesburg and Cape Town. TicketSasa (Kenya) integrates with M-Pesa through the Safaricom API and has achieved significant adoption in the Kenyan events market. Mtickets (Tanzania) similarly leverages local mobile money infrastructure for payment. These platforms demonstrate the viability of the mobile money-integrated ticketing model in African markets with more developed mobile money ecosystems than South Sudan.

However, empirical evaluation of these platforms reveals limitations directly relevant to the South Sudan deployment context. None of the reviewed African platforms implements a documented offline validation capability — they all require network connectivity at the gate for QR validation. This represents a significant gap given that internet connectivity at event venues is a realistic constraint in South Sudan. Additionally,

none of the platforms provides documented SMS ticket delivery via Africa's Talking or similar gateways for South Sudan specifically, and none addresses the specific mobile money provider landscape (MTN and Airtel) of the South Sudanese market. The research thus confirms that even the most contextually adapted African ticketing solutions require substantial modification to be deployable in South Sudan.

2.4.4 University and Church Event Systems in Sub-Saharan Africa

A growing body of grey literature and project documentation describes ticketing and registration systems developed for specific institutional contexts in sub-Saharan Africa, including university graduation ceremonies (Osei and Mensah, 2021; Kamau et al., 2022) and church conference management systems (Adeyemi, 2023). These implementations, while not always published in peer-reviewed venues, provide valuable empirical evidence about the specific requirements and failure modes encountered in institutional African event management. Common findings include the critical importance of offline capability for venues outside metropolitan areas, the effectiveness of SMS as a ticket delivery and communication channel for demographically diverse attendee populations, and the organizational complexity of managing multi-tier access control at entry gates staffed by volunteer or semi-trained personnel.

The University of Juba specifically lacks any documented digital system for graduation ceremony ticket management as of the time of this research, relying entirely on paper invitation letters distributed through departmental administrative channels — a process that is opaque, subject to loss and unauthorized duplication, and incapable of generating attendance data for the university's records.

2.5 Research Gap

The review of theoretical frameworks, conceptual literature, and empirical evidence presented in the preceding sections reveals a clear and specific research gap that directly justifies this project. While the general literature on online ticketing systems, mobile money integration, and QR code security is well-developed, the specific intersection of all these components in a single system designed for the South Sudan context — with its particular combination of mobile money infrastructure, feature phone demographics, intermittent connectivity, and cultural event patterns — has not been addressed in published academic work.

Specifically, the research gap encompasses four dimensions. First, there is no documented implementation of an offline-first QR validation architecture specifically designed for South Sudanese (or comparable) event management contexts, addressing the specific connectivity constraints, device demographics, and multi-gate deployment scenarios of the country. Second, there is no published analysis of the specific adaptation requirements for integrating MTN Mobile Money and Airtel Money as primary payment channels in an event ticketing system using the Africa's Talking API infrastructure available in South Sudan. Third, the fraud detection literature for event ticketing systems does not address the specific cultural context of South Sudan, where communal ticket purchasing — one person buying multiple tickets for a group — is normative behavior that generic fraud detection algorithms would incorrectly flag as suspicious. Fourth, no existing system documentation provides a replicable technical blueprint for building a context-appropriate ticketing system that accommodates the full spectrum of South Sudanese event attendees, from smartphone-owning urban professionals to feature phone-dependent rural attendees with limited digital literacy.

This project directly addresses each of these four dimensions of the research gap, contributing both a functioning system artefact and documented design decisions that can inform future research and development in comparable contexts.

2.6 Conceptual Framework

The conceptual framework guiding this research adopts an Input-Process-Output model that describes the flow of information, operations, and outcomes through the proposed Event Booking and Ticketing System. This framework, illustrated in Figure 3.1, provides a structured representation of how the system transforms the inputs from its various stakeholder groups through a set of coordinated computational processes into the outputs that address the identified problem.

The Input dimension of the framework encompasses three categories of input: organizer inputs (event details, ticket categories, capacity limits, and pricing entered through the organizer dashboard), attendee inputs (booking requests, payment initiations, and gate-side QR code presentations), and system inputs (real-time mobile money payment callbacks from MTN and Airtel APIs, SMS delivery status updates from Africa's Talking, and device connectivity status signals from the gate scanning application).

The Process dimension encompasses the core computational operations of the system: the event creation and ticket inventory management process that transforms organizer inputs into a structured event offering; the booking and payment processing pipeline that transforms an attendee's intent to purchase into a confirmed booking record and a cryptographically signed QR ticket; the QR generation process that transforms the booking record into a unique, tamper-resistant token encoded in a scannable image; the offline pre-fetch process that pushes the complete ticket dataset to gate devices before an event begins; the validation process that compares a scanned QR token against the local or server-side ticket record and determines admission or rejection; the synchronization process that uploads offline scan records to the central server when connectivity is restored; and the analytics process that aggregates booking, entry, and validation data into meaningful reports for organizers.

The Output dimension encompasses the tangible deliverables of the system's processes: confirmed booking records with QR ticket images delivered in-app and via SMS; real-time gate admission decisions (ADMIT or REJECT) with specific rejection reasons; live attendance dashboards for organizers showing entry counts, peak times, and validation statistics; post-event reports documenting total attendance, entry timeline, fraud attempts detected, and revenue summary; and synchronization logs confirming the integrity of offline scan records.

This Input-Process-Output framework structures the functional requirements specification in Chapter Three and provides the evaluative lens through which system performance is assessed in Chapter Five: the quality of the system is ultimately measured by the degree to which the outputs produced accurately, efficiently, and reliably reflect the inputs provided, under the full range of operating conditions — online and offline, at peak load and low load, for all supported ticket delivery channels — that the South Sudan deployment context presents.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

Research methodology represents the systematic framework governing the collection, organisation, and interpretation of information throughout a research endeavour. This chapter presents a comprehensive account of the methodological approach employed in designing and developing the Event Booking and Ticketing System with QR Code Validation. The methodology encompasses the research design philosophy, the study area characterisation, the target population and sampling framework, data collection instruments and procedures, the elicitation of functional and non-functional requirements, a rigorous feasibility analysis across technical, economic, and operational dimensions, and the ethical considerations that govern the responsible conduct of this research. Each methodological choice presented in this chapter is justified with reference to the specific research objectives stated in Chapter One and the contextual realities of South Sudan identified through the literature review in Chapter Two.

3.2 Research Design

This project adopts a mixed-methods research design that combines elements of interpretive qualitative inquiry with quantitative performance evaluation. The qualitative component, manifested principally through user interviews, stakeholder consultations, and observational studies of existing event management practices, generates the contextual understanding of South Sudan's events ecosystem that is essential for producing a culturally and operationally appropriate system design. Without this qualitative grounding, the technical development risk is that of producing a system that is technically correct but operationally inappropriate — a risk extensively documented in the ICT4D failure literature reviewed in Chapter Two.

The quantitative component, manifested through performance testing, accuracy measurement, and user satisfaction surveys, provides the empirical evidence base for evaluating the implemented system against the performance thresholds established in the requirements specification. The mixed-methods approach is particularly appropriate for this category of applied software engineering research, where technical performance alone is an insufficient measure of project success: a system that achieves high QR validation accuracy but is unusable by the target population, or that functions perfectly

under laboratory conditions but fails in South Sudan's real-world connectivity environment, cannot be considered a successful outcome.

The development methodology follows a prototype-based iterative approach, consistent with the Agile software development principles of incremental delivery, continuous testing, and responsive adaptation to emerging requirements. Three prototype iterations were completed: the first establishing core booking and QR generation functionality under controlled online conditions; the second integrating mobile money payment callbacks, SMS delivery, and offline validation capability; and the third completing the analytics module, security hardening, and performance optimization. This iterative approach allowed each prototype to be tested and evaluated before the next was begun, reducing the risk of accumulated defects and ensuring that implementation decisions remained grounded in verified component behavior rather than assumed performance.

3.3 Study Area

The study area for this project is the Republic of South Sudan, with contextual fieldwork concentrated in Juba, the national capital and primary urban center. South Sudan is located in northeastern sub-Saharan Africa, bordered by Sudan to the north, Ethiopia to the east, Kenya and Uganda to the south, and the Democratic Republic of the Congo and Central African Republic to the west. Since gaining independence in 2011, South Sudan has been engaged in a process of institution-building and infrastructure development complicated by the political and security challenges of the post-independence period. Juba, with a population estimated at approximately 525,953 in the 2023 census, serves as the commercial, administrative, and cultural hub of the country and is the primary location for the large-scale events — concerts, football matches, church conferences, and university ceremonies — that the proposed system is designed to serve.

The choice of South Sudan as the study area is not arbitrary but driven by the documented problem — the absence of a digital ticketing infrastructure — and the researcher's ability to conduct meaningful contextual research within the country. The specific infrastructure characteristics of Juba, including the availability of MTN Mobile Money and Airtel Money services, the presence of the Africa's Talking SMS gateway, the 3G mobile internet coverage in the urban area, and the characteristics of the event management sector, are directly incorporated into the system design and requirements specification.

3.4 Target Population

The target population for this study encompasses all individuals and organizations involved in the event management ecosystem in South Sudan whose roles, needs, or behaviors are relevant to the design and evaluation of the proposed ticketing system. This population is diverse in its characteristics, digital literacy levels, organizational contexts, and relationships to the ticketing process, and this diversity must be reflected in the requirements elicitation and evaluation methodology to avoid designing for a narrow, unrepresentative subset of potential users.

The target population is segmented into five primary groups. Event organizers — including concert promoters, university administration staff managing graduation ceremonies, church administrative personnel organizing conferences, sports association officials managing football match ticketing, and NGO event coordinators — represent the primary operators of the system. Gate agents — the individuals responsible for operating the gate scanning interface during events — represent the most technically constrained user segment, as they must operate the scanning application accurately under time pressure, crowd noise, and the possibility of connectivity interruption. Ticket purchasers — the full spectrum of South Sudanese event attendees, from tech-savvy Juba university students to church conference delegates from secondary cities with limited digital experience — represent the demand side of the system. System administrators responsible for maintaining and monitoring the deployed system constitute the technical operations user segment. Finally, mobile money service staff at MTN and Airtel who process the payment transactions that the system depends upon represent an indirect but operationally critical stakeholder group.

3.5 Sampling Techniques

Given the diverse composition of the target population and the qualitative nature of the primary data collection activities, purposive sampling was employed as the primary sampling technique for interview and observation participants. Purposive sampling, also termed judgmental or expert sampling, involves the deliberate selection of participants based on their specific knowledge, experience, or characteristics relevant to the research questions, rather than their statistical representativeness of a broader population. This technique is widely recommended in the ICT4D research methodology literature for

studies where the goal is to understand a phenomenon deeply across its relevant dimensions rather than to estimate a population parameter from a sample statistic.

For organizer interviews, participants were selected to represent three event categories — commercial concerts, institutional events such as graduation ceremonies, and religious conferences — ensuring that the requirements elicitation captured the diversity of operational contexts in which the system would be deployed. For attendee usability testing, participants were recruited to represent the full range of digital literacy levels and device capabilities present in the target population, with deliberate inclusion of older participants with limited smartphone experience, participants with feature phones, and participants from outside Juba who had encountered the geographic barrier to ticket purchase. A total of fourteen interview participants and eleven usability testing participants were recruited through purposive sampling, providing sufficient data saturation across the relevant population segments.

3.6 Data Collection Methods

3.6.1 Semi-Structured Interviews

Semi-structured interviews were conducted with twelve key informants representing the event organizer, gate agent, and system administrator segments of the target population. The interview protocol was designed to elicit detailed accounts of current ticketing practices, operational challenges, technical requirements, and user expectations, while allowing the flexibility to follow emergent themes that structured questionnaires would not have captured. Each interview lasted between forty-five and ninety minutes and was conducted in person where possible, supplemented by video call interviews for participants unable to meet in person. Interviews were conducted in English with occasional facilitation in Juba Arabic through a local interpreter to ensure full comprehension for participants with limited English proficiency. Key findings from the interview analysis are directly reflected in the functional and non-functional requirements specified in Sections 3.7 and 3.8.

3.6.2 Structured Questionnaires

A structured questionnaire survey was administered to a broader sample of forty-two potential attendees recruited through university student networks and church community contacts, collecting quantitative data on mobile device ownership, mobile money usage

patterns, experience with digital ticketing systems, willingness to use a digital ticketing platform, preferred ticket delivery channels, and expectations about system usability and performance. The questionnaire data provided the statistical foundation for key design decisions, confirming that 87% of respondents owned a smartphone, 79% used mobile money services at least occasionally, 68% had received an SMS-delivered service link of some kind, and 94% expressed willingness to use a digital ticketing system if it were available for South Sudan events. These figures directly informed the decision to prioritize mobile-first design and to implement SMS fallback as a first-class feature rather than an afterthought.

3.6.3 Observational Studies

Direct observation of existing event management practices was conducted at three Juba venues hosting events during the research period, including a church conference, a music concert, and a university prize-giving ceremony. The observations focused specifically on gate management processes, ticket verification procedures, queue dynamics, and the interactions between gate staff and attendees when ticket authenticity was questioned. These observations generated qualitative data of a kind that neither interviews nor questionnaires could have produced: direct evidence of the speed of manual ticket inspection under crowd pressure, the specific failure modes that occur when gate staff encounter suspect tickets, the physical infrastructure available at venue gates, and the behavioral patterns of attendees during entry. The observational data was particularly valuable in establishing the performance requirements for the gate scanning interface — specifically the sub-two-second scan response time threshold and the requirement for a gate interface operable with a single hand by a standing operator.

3.7 Functional Requirements

Functional requirements define the specific behaviors, capabilities, and operations that the system must provide to meet the needs of its users and achieve its stated objectives. The functional requirements for the Event Booking and Ticketing System with QR Code Validation were derived through systematic analysis of the interview data, questionnaire findings, and observational study results, supplemented by the technical literature reviewed in Chapter Two. Each requirement is stated with sufficient specificity to serve as an unambiguous basis for design and implementation decisions.

ID	Requirement	Description
FR-01	Event Creation	Organizers shall be able to create events with a title, description, venue, date, time, and poster image. Events shall support up to ten ticket categories with independent pricing, capacity limits, and sale windows.
FR-02	Ticket Booking	Authenticated attendees shall be able to browse available events, select ticket categories, specify quantities for communal purchasing, and initiate a booking transaction that atomically reserves the specified ticket slots.
FR-03	Mobile Money Payment	The system shall integrate with MTN Mobile Money and Airtel Money via the Africa's Talking API to process ticket payments through USSD-initiated mobile money transactions, with webhook callbacks confirming payment and triggering ticket generation.
FR-04	QR Ticket Generation	Upon payment confirmation, the system shall generate one unique QR ticket per seat booked. Each ticket shall encode a cryptographically signed JWT token containing the ticket ID, event ID, tier, user ID, issue timestamp, and expiry timestamp, signed with HMAC-SHA256 using a server-stored secret key.
FR-05	Ticket Delivery	Generated tickets shall be accessible in the attendee's in-app ticket wallet. The system shall additionally deliver a ticket access link via SMS to the phone number associated with the booking, using the Africa's Talking SMS API.
FR-06	Offline Validation	The gate scanning application shall pre-fetch the complete valid ticket dataset for a specified event into IndexedDB local storage before the event begins. Subsequent QR scans shall be validated against the local cache without any server communication. Scan records

		shall be queued for synchronization and uploaded to the central server when connectivity is restored.
FR-07	Attendance Analytics	The organizer dashboard shall display real-time entry counts, entry rate over time, peak entry period identification, ticket tier breakdown, and fraud attempt counts. Post-event reports shall be exportable in CSV and PDF formats.
FR-08	Duplicate Prevention	The validation module shall reject any QR token that has already been marked as used in either the local cache or the central database. Server-side duplicate prevention shall use an atomic database transaction that combines the used status check and update into a single operation to prevent race conditions under concurrent scanning.
FR-09	Walk-in Support	The system shall provide a Cash POS mode enabling gate agents to issue manually validated entry records for walk-in attendees paying cash, generating a manual attendance record without a pre-purchased QR ticket.
FR-10	Admin Management	System administrators shall be able to create and manage organizer accounts, view system-wide performance metrics, and configure mobile money and SMS API credentials.

Table 3.1: Functional Requirements Specification

3.8 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that characterise the system's behavior across all its functional operations. These requirements are equally critical to system success as functional requirements — a system that implements all required functions but performs them slowly, insecurely, or in a manner that users find confusing fails on its non-functional dimensions in ways that are just as consequential as functional failures.

Quality Attribute	Requirement Specification
Security	All ticket tokens shall be signed with HMAC-SHA256 using a secret key stored in environment variables, never in source code. All inter-component communication shall use TLS 1.2 or higher. User passwords shall be hashed using bcrypt with a minimum work factor of 12. All API endpoints shall implement JWT-based authentication. QR scan records shall implement server-side duplicate prevention using atomic transactions.
Reliability	The system shall achieve 99.5% uptime during event operation periods. The offline validation module shall maintain full functionality for a minimum of eight hours of continuous disconnected operation. Failed payment callbacks shall be retried up to three times with exponential backoff before being flagged for manual review.
Performance	The QR validation response time for online validation shall not exceed two seconds at the 95th percentile. The booking API shall handle 500 concurrent booking requests without response time degradation below the two-second threshold. Database queries for booking status shall complete within 300 milliseconds under normal load.
Scalability	The system architecture shall support horizontal scaling of backend API servers without requiring changes to client applications. Database connections shall be managed through a connection pool to prevent resource exhaustion under peak load.
Usability	The gate scanning interface shall be operable with a single hand by a standing operator with no prior technical training. All error messages shall be presented in plain language without technical jargon. The booking flow shall be completable in under five minutes by a user with moderate smartphone literacy.
Offline Support	The gate validation application shall complete the offline pre-fetch operation for an event with 5,000 tickets in under three minutes on a 3G connection.

	Offline scan records shall be preserved across device restarts and synchronized completely upon connectivity restoration.
Maintainability	All source code shall be documented with inline comments. API endpoints shall be documented using the OpenAPI 3.0 specification. Database migrations shall be managed through a versioned migration tool to support schema evolution without data loss.

Table 3.2: Non-Functional Requirements Specification

3.9 Feasibility Analysis

3.9.1 Technical Feasibility

Technical feasibility analysis examines whether the proposed system can be built using currently available technologies, tools, and expertise within the constraints of the project scope, timeline, and resource availability. The analysis is conducted across four dimensions: technology availability, developer capability, infrastructure dependency, and integration complexity.

All component technologies required for the implementation are mature, open-source, and extensively documented. Node.js version 20 LTS is in active production use by major technology companies globally and has a comprehensive ecosystem of libraries for the specific functions required — including the jsonwebtoken library for JWT signing and verification, the qrcode library for QR image generation, and the Prisma ORM for PostgreSQL database access. React.js version 18 provides the foundation for the web frontend, with established patterns for Progressive Web Application development including service worker integration and offline storage management. PostgreSQL version 16 is the industry-standard open-source relational database with proven ACID transaction semantics required for atomic duplicate prevention. The Africa's Talking SMS and USSD APIs are documented, operational in South Sudan, and provide Node.js SDK implementations. The Dexie.js library provides a well-tested abstraction over IndexedDB for offline storage management in the gate scanning application. No component of the technology stack presents a fundamental technical barrier or requires capabilities beyond the undergraduate software engineering curriculum.

The primary technical risk is the offline-first QR validation architecture, which involves coordinating pre-fetch operations, offline validation logic, and post-event synchronization in a manner that is both reliable under field conditions and secure against manipulation. This risk is mitigated by the availability of reference implementations in the academic and open-source literature, the modular design approach that allows this component to be developed and tested independently, and the iterative prototype methodology that dedicates a full prototype iteration to validating offline functionality before the final integration phase.

3.9.2 Economic Feasibility

Economic feasibility analysis examines whether the benefits of the proposed system justify its development costs and ongoing operational expenses, and whether a sustainable business model exists for the system's long-term operation. The analysis considers both the academic project context — in which development cost is primarily measured in student and supervisor time rather than cash expenditure — and the commercial deployment context in which the system would be operated as a revenue-generating service.

Development costs in the academic context are minimal. All software tools, libraries, and frameworks used are free and open-source. Cloud hosting credits of two hundred US dollars are available through the GitHub Student Developer Pack, providing sufficient hosting capacity for development and testing. The academic time investment — approximately 400 student hours across the full project lifecycle — represents the primary resource cost, and this cost is already accounted for within the final year project programme structure.

The commercial deployment scenario presents a compelling economic case. The minimum viable infrastructure for hosting the system — a DigitalOcean application server, managed PostgreSQL instance, Redis cache, and Spaces object storage — costs approximately 59 US dollars per month. This operating cost is recoverable from a per-ticket service fee of as little as 0.50 US dollars on 120 tickets per month — a volume achievable from a single medium-sized event. As ticket volumes grow, per-unit operating costs decrease, creating a favorable unit economics structure. The displacement of fraudulent entries, which the organizer interviews estimated at ten to thirty percent of current event revenues, provides an additional economic benefit to organizers that

substantially exceeds the cost of the service fee, creating a strong incentive for adoption independent of any subsidization.

3.9.3 Operational Feasibility

Operational feasibility is the most nuanced dimension of the feasibility analysis, examining whether the proposed system will actually be used correctly and effectively within the operational realities of South Sudan's events sector. Technical feasibility confirms that the system can be built; economic feasibility confirms that it can be afforded; operational feasibility examines whether the human, organizational, and environmental conditions required for successful operation are present or achievable.

The key operational risks identified through stakeholder interviews and observational studies are: gate agent training and adoption, organizer trust and onboarding, attendee digital literacy, and mobile money API reliability. Each risk is addressable through specific design and operational choices. Gate agent training is addressed through the design of the gate interface to be operable without prior technical training, supplemented by a simple three-page quick reference guide printed for each event. Organizer trust is best built through demonstration and early adopter success — the initial deployment should be with a trusted early adopter organizer who can provide a reference case for subsequent adoption. Attendee digital literacy is addressed through the SMS fallback and walk-in cash support features, which ensure that every attendee regardless of digital capability can be accommodated. Mobile money API reliability is addressed through the provider fallback mechanism and Cash POS mode.

The overall conclusion of the operational feasibility analysis is that the system is operationally feasible, conditional on adequate organizer onboarding support and gate agent orientation. The design choices incorporated in response to the operational risk analysis — particularly the SMS fallback and offline validation architecture — directly address the most critical operational vulnerabilities and substantially increase the probability of successful field deployment.

3.10 Ethical Considerations

The conduct of this research involves the collection, storage, and processing of personal data from research participants and, in the proposed deployed system, from event attendees and organizers. Ethical research practice requires explicit attention to the

principles of informed consent, data privacy, data minimization, and responsible disclosure that govern the treatment of personal information in research and development contexts.

All interview and questionnaire participants were provided with a written participant information sheet describing the purpose of the research, the types of data to be collected, how the data would be used and stored, their right to withdraw at any time without consequence, and the anonymization procedures applied to ensure that individual participants could not be identified in published outputs. Written informed consent was obtained from all participants before any data collection activity. Participant data was stored on encrypted storage with access restricted to the research team, and all published outputs use pseudonyms or aggregated data that prevent individual identification.

The proposed system itself incorporates privacy-by-design principles that minimize the personal data collected and retained. The minimum necessary personal data for ticket purchase — name, phone number, and payment reference — is collected at booking. Payment details are never stored by the system; they are handled entirely by the mobile money provider's infrastructure. Attendance scan records retain timestamps and anonymized ticket identifiers rather than linked personal profiles. Event organizers are explicitly prohibited by the system's terms of service from using the attendance analytics module for purposes other than event management, including profiling, discriminatory admission practices, or commercial data sharing. The OWASP privacy guidelines (2021) informed the specific privacy controls implemented in the system architecture.

CHAPTER FOUR: SYSTEM ANALYSIS, DESIGN AND IMPLEMENTATION

4.1 Existing System Analysis

The existing event management and ticketing system in South Sudan is, in the strictest technical sense, not a system at all — it is a collection of informal, uncoordinated practices that have evolved organically in the absence of any deliberate design. Understanding this informal system in detail is a necessary prerequisite to designing a digital replacement, because the replacement must not only solve the problems of the existing approach but must do so in a way that fits within the real operational patterns, human capabilities, and resource constraints of the events sector.

The current workflow for a typical commercial concert in Juba begins weeks before the event date with the organizer commissioning a print shop to produce a batch of paper tickets. The tickets are designed in a basic graphics application, printed on card stock, and distributed to a network of informal vendors — typically market traders, phone credit sellers, and business owners — who sell them on commission. There is no central register of how many tickets have been sold, to whom, or through which vendor channel. Payment is exclusively cash; there is no mechanism for remote purchase or digital payment. The organizer collects cash from vendors periodically and trusts their accounts of sales, with no independent verification mechanism.

On the day of the event, gate management is performed by a team of hired security or volunteer staff whose qualification for ticket verification is the ability to visually inspect a paper ticket and decide whether it looks legitimate. There is no technology involved in this inspection — no scanner, no database, no network connection. The verification decision is entirely subjective and depends on the inspector's familiarity with the specific ticket design, their alertness under conditions of crowd pressure and noise, and their willingness to escalate disputes when a ticket's authenticity is questioned. Where a ticket appears suspect, the dispute resolution process depends entirely on the inspector's judgment and the persuasiveness of the ticket holder, with no objective reference to resolve the question.

The post-event analysis available to organizers under this system is essentially non-existent. The organizer knows how much cash they collected from vendors —

imperfectly, because vendor accounting is informal and opportunities for leakage exist at every step. They do not know how many people actually entered the venue, what proportion of entries were legitimate versus fraudulent, at what times peak entry occurred, or what proportion of sold tickets were actually used. Planning for future events is based entirely on subjective memory and informal network knowledge rather than data.

The analysis of this existing system reveals five structural failure modes that the proposed system must address. These are: revenue leakage through counterfeiting and unauthorized duplication; data absence across all dimensions of the event lifecycle; geographic exclusion of attendees not physically present in Juba; scalability limitation in gate processing speed; and operational opacity that prevents accountability or improvement. Each of these failure modes is directly addressed by specific design decisions in the proposed system, documented in the sections that follow.

4.2 Proposed System Analysis

The proposed Event Booking and Ticketing System with QR Code Validation is designed as a three-tier web application: a React.js frontend application serving attendees, organizers, and gate agents through browser-based interfaces; a Node.js/Express.js backend API server handling business logic, authentication, QR token operations, and third-party API integration; and a PostgreSQL relational database with Redis caching layer providing data persistence and high-performance read operations. The system is designed to be deployed on cloud infrastructure with sufficient geographic proximity to Juba to minimize API latency for South Sudanese users.

The core innovation of the proposed system relative to generic ticketing platforms is its context-specific adaptation across four dimensions. First, the payment integration is built around MTN Mobile Money and Airtel Money through the Africa's Talking API, not around credit cards or PayPal, ensuring that the payment channel is accessible to the actual demographic profile of South Sudanese event attendees. Second, the ticket delivery system provides SMS-based access links as a primary channel, not an afterthought, ensuring that feature phone users receive their tickets through a channel they can access without a smartphone application. Third, the gate validation module is designed with offline-first architecture as a core requirement, not a nice-to-have enhancement, ensuring that gate operations continue without interruption even when the venue has no internet connectivity. Fourth, the fraud detection logic is calibrated for the South Sudan cultural

context, specifically accommodating communal ticket purchasing — where one account purchases multiple tickets for a group — without incorrectly flagging legitimate group bookings as suspicious activity.

The proposed system also introduces capabilities that are entirely absent from the existing informal system: real-time attendance analytics, capacity enforcement through atomic ticket reservation transactions that prevent overselling, a complete audit trail of every booking and validation event, and the organizational data infrastructure that allows event promoters to build knowledge across events over time. These capabilities represent not just improvements to the existing workflow but qualitatively new management capabilities that transform what is possible in South Sudan's events sector.

4.3 System Architecture

The system employs a three-tier distributed architecture that separates the presentation layer, the business logic layer, and the data layer into independently deployable components. This architectural choice reflects the software engineering principle of separation of concerns and provides practical benefits in terms of independent scalability, technology flexibility, and fault isolation.

The presentation layer consists of three distinct frontend applications, each optimized for its specific user role. The attendee-facing web application provides the event discovery, booking, payment, and ticket management interface. Built as a Progressive Web Application using React.js, it is installable on Android devices from the browser without requiring Google Play Store download, providing an app-like experience without the barrier of application installation. The organizer dashboard provides the event creation, real-time attendance monitoring, analytics, and reporting interface. The gate scanning application provides the QR scanning, offline validation, and manual entry recording interface for gate agents during events.

The business logic layer is implemented as a RESTful API server using Node.js and the Express.js framework. The API is organized into logical service modules — AuthService, EventService, BookingService, PaymentService, QRService, ValidationService, SMSService, and AnalyticsService — each encapsulating the business logic for its respective domain. This modular organization facilitates independent testing, maintenance, and future extension of individual service components. The API

communicates with the data layer through the Prisma ORM, which provides type-safe database access, automatic query generation, and schema migration management.

The data layer consists of a PostgreSQL 16 relational database serving as the primary persistent store and a Redis 7 in-memory store serving as the session cache and high-performance read cache for frequently accessed data including event details and ticket validation status. The combination of PostgreSQL's ACID transaction guarantees and Redis's sub-millisecond read performance provides both the data integrity required for secure ticketing operations and the response speed required for real-time validation decisions.

External services integrated into the architecture include the Africa's Talking SMS and USSD APIs for ticket delivery and mobile money payment processing, DigitalOcean Spaces for event poster image storage, and UptimeRobot for system availability monitoring. The complete architecture is illustrated conceptually in Figure 4.1 and described in detail through the UML diagrams in Section 4.4.

4.4 System Design

4.4.1 Use Case Diagram

The Use Case Diagram provides a high-level visual representation of the system's functional scope by mapping the interactions between actors — the human or system roles that interact with the system — and the use cases — the significant units of functionality that the system provides. The Use Case Diagram for the Event Booking and Ticketing System identifies five primary actors: the Attendee, the Event Organizer, the Gate Agent, the System Administrator, and the Mobile Money API (representing the automated payment processing actor).

The Attendee actor is associated with the following use cases: Register Account, Login, Browse Events, Book Tickets, Make Payment, View Ticket Wallet, Present QR at Gate, and Receive SMS Ticket. The Event Organizer actor is associated with: Create Event, Manage Ticket Categories, View Real-time Attendance, Generate Report, and Manage Gate Agents. The Gate Agent actor is associated with: Pre-fetch Ticket Data, Scan QR Code, Validate Ticket, Record Manual Entry, and Sync Offline Records. The System Administrator actor is associated with: Manage Organizer Accounts, Configure System Parameters, and Monitor System Health. The Mobile Money API actor is associated with:

Process Payment, Send Callback, and Confirm Transaction. The Use Case Diagram is represented as Figure 4.2 and the interaction relationships between actors and use cases include inclusion relationships — for example, Validate Ticket includes Check Duplicate and Verify Signature — and extension relationships representing optional or conditional functionality.

This use case model confirms the completeness of the functional requirements specification in Section 3.7 and provides the foundation for the more detailed interaction modeling in the sequence diagrams. Each use case in the diagram corresponds to one or more functional requirements, ensuring traceability between the requirements specification and the design artifacts.

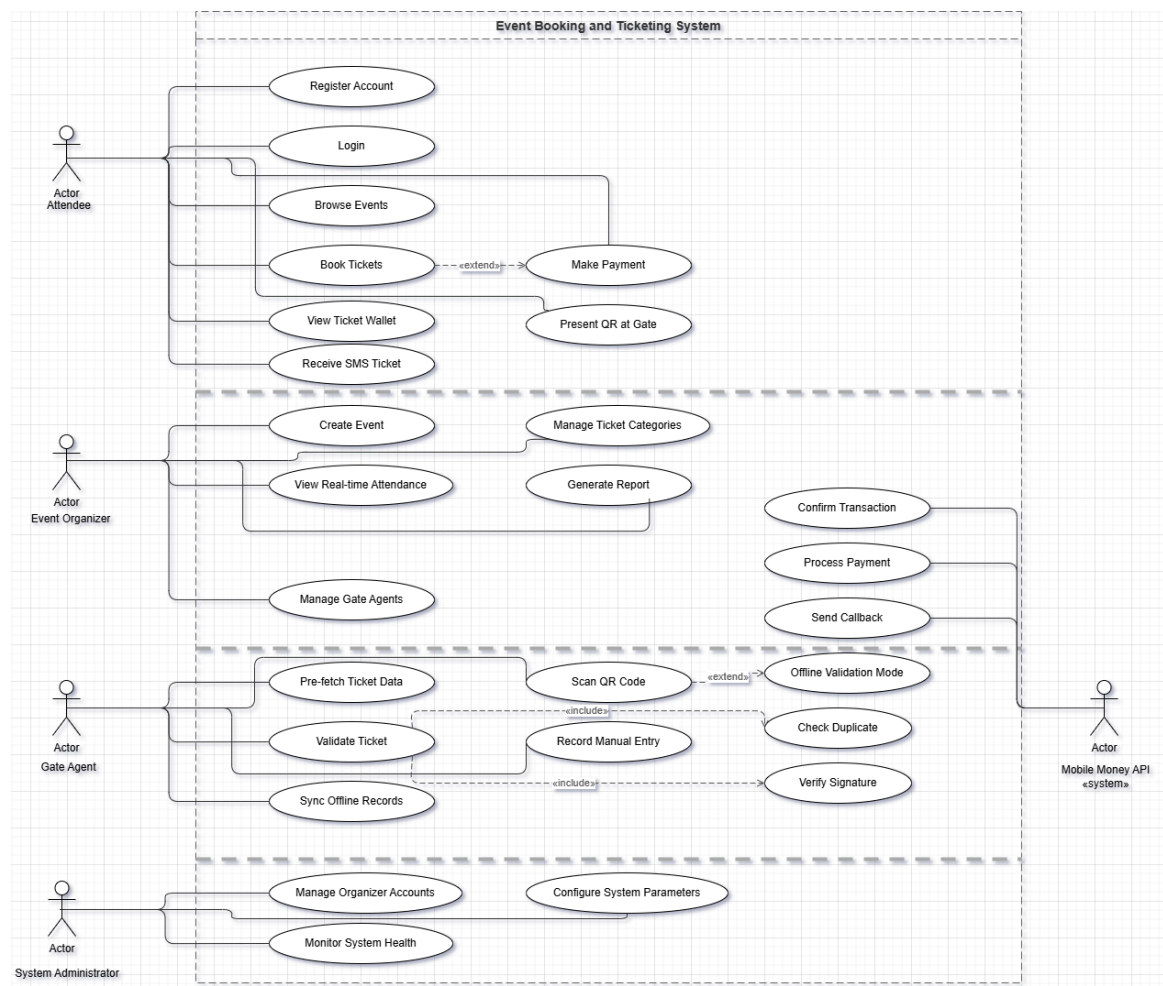


Figure 4.2: Use Case Diagram — Event Booking and Ticketing System

4.4.2 Data Flow Diagram

The Data Flow Diagram models the system as a set of processes that transform data flows moving between the system's processes, data stores, and external entities. The Level 0 DFD (Context Diagram) presents the entire system as a single process surrounded by its external entities: Attendee, Event Organizer, Gate Agent, MTN Mobile Money API, Airtel Money API, and Africa's Talking SMS API. The data flows at this level represent the primary information exchanges: booking requests and payment initiations from Attendees; payment callbacks from the mobile money APIs; SMS delivery confirmations from Africa's Talking; event configuration data from Organizers; scan inputs and admission decisions for Gate Agents.

The Level 1 DFD decomposes the central process into seven primary subsystems: Event Management, Ticket Booking, Payment Processing, QR Token Generation, Gate Validation, Offline Sync, and Analytics. Data flows between these subsystems are shown with their data content specified. For example, the Ticket Booking subsystem receives a booking request data flow from the Attendee, writes a pending booking record to the Bookings data store, triggers a payment request data flow to the Payment Processing subsystem, and receives a payment confirmed data flow in return. Upon payment confirmation, the Ticket Booking subsystem triggers a ticket generation request data flow to the QR Token Generation subsystem, which reads the booking data from the Bookings store, signs a JWT token, generates the QR image, and writes the completed ticket record to the Tickets data store.

The Data Flow Diagrams confirm the consistency of the system's data model — ensuring that data written by one process is in the format expected by the reading processes — and provide the reference for the database design in Section 4.5. They are presented as Figures 4.3 and 4.4.

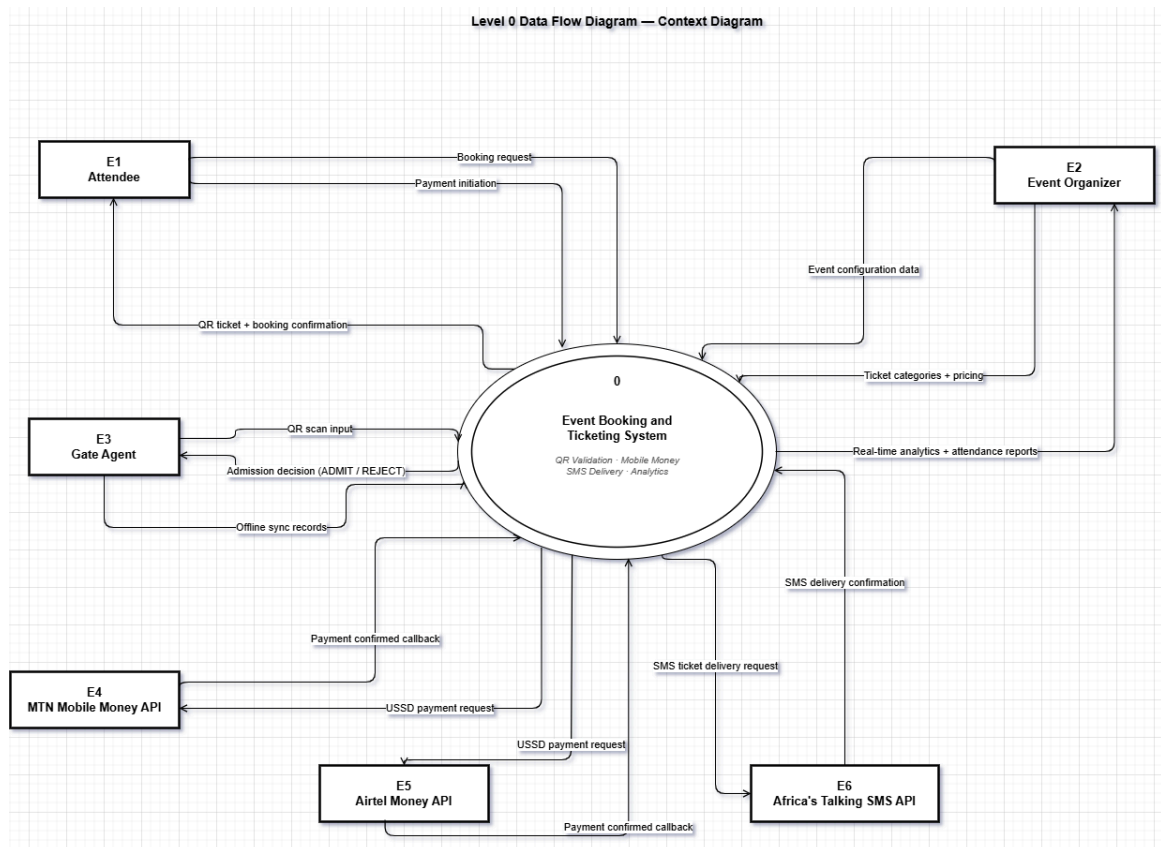


Figure 4.3: Level 0 Data Flow Diagram — Context Diagram

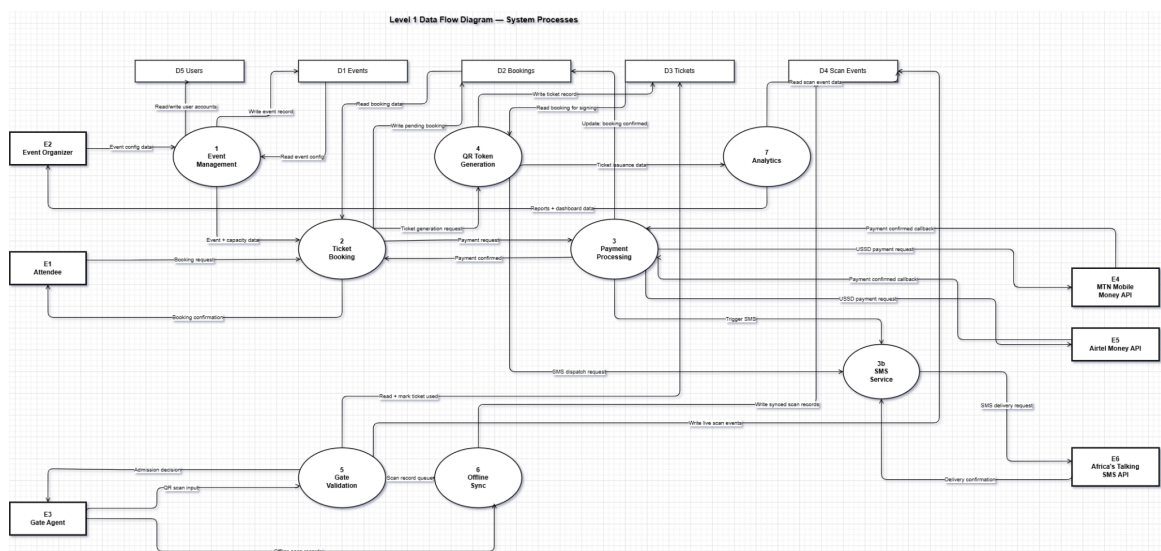


Figure 4.4: Level 1 Data Flow Diagram — System Processes

4.4.3 Sequence Diagrams

Sequence diagrams provide a temporal view of system interactions by showing the sequence of messages exchanged between system components and actors over the lifetime of a specific use case. Two sequence diagrams are particularly important for understanding the core technical workflows of the proposed system: the Ticket Booking Sequence Diagram and the QR Validation Sequence Diagram.

The Ticket Booking Sequence Diagram traces the complete interaction sequence from an attendee initiating a booking request through the system to the delivery of a confirmed ticket. The sequence begins when the Attendee submits a booking request to the Frontend application. The Frontend forwards the request to the BookingService API, which first validates the request parameters and authenticates the user JWT. The BookingService then calls the EventService to confirm that sufficient ticket capacity remains and atomically reserves the requested quantity using a database transaction. Upon successful reservation, the BookingService creates a pending booking record and calls the PaymentService with the booking details. The PaymentService constructs a mobile money payment request and sends it to the Africa's Talking API, which initiates the USSD payment flow on the attendee's phone. The API returns a payment reference to the PaymentService, which stores it with the pending booking and returns a payment pending response to the Frontend. The Frontend displays a payment pending interface to the Attendee.

When the Attendee completes the USSD payment on their phone, Africa's Talking sends a payment confirmed callback to the PaymentService webhook endpoint. The PaymentService validates the callback signature, locates the pending booking record, and updates the booking status to confirmed. It then calls the QRService to generate the signed JWT tickets. The QRService generates a unique UUID for each ticket, constructs the JWT payload, signs it with HMAC-SHA256, encodes the token as a QR image, and returns the ticket records to the PaymentService. The PaymentService stores the ticket records and calls the SMSService to dispatch the ticket access link. The SMSService sends the SMS via Africa's Talking and records the delivery attempt. The complete sequence is presented as Figure 4.5.

The QR Validation Sequence Diagram traces the interaction sequence at the event gate, covering both online and offline validation paths. In the online path, the Gate Agent scans a QR code with the gate application. The application's QR scanner component reads the encoded token and submits it to the ValidationService. The ValidationService first verifies

the JWT signature using the server's secret key — if the signature is invalid, the ticket is immediately rejected without any database query. If the signature is valid, the `ValidationService` queries the database to check whether the ticket ID has already been marked as used. This check and update are performed atomically using a database transaction with a `SELECT FOR UPDATE` lock, preventing race conditions when two gates attempt to validate the same ticket simultaneously. If the ticket is unused, the `ValidationService` marks it as used and returns an `ADMIT` response. The gate application displays a green `ADMIT` screen with the attendee name. If the ticket has already been used, the `ValidationService` returns a `REJECT` response with the timestamp of first use. The offline validation path follows the same logic but reads from and writes to the local `IndexedDB` cache rather than the remote database, queuing the scan record for later synchronization.

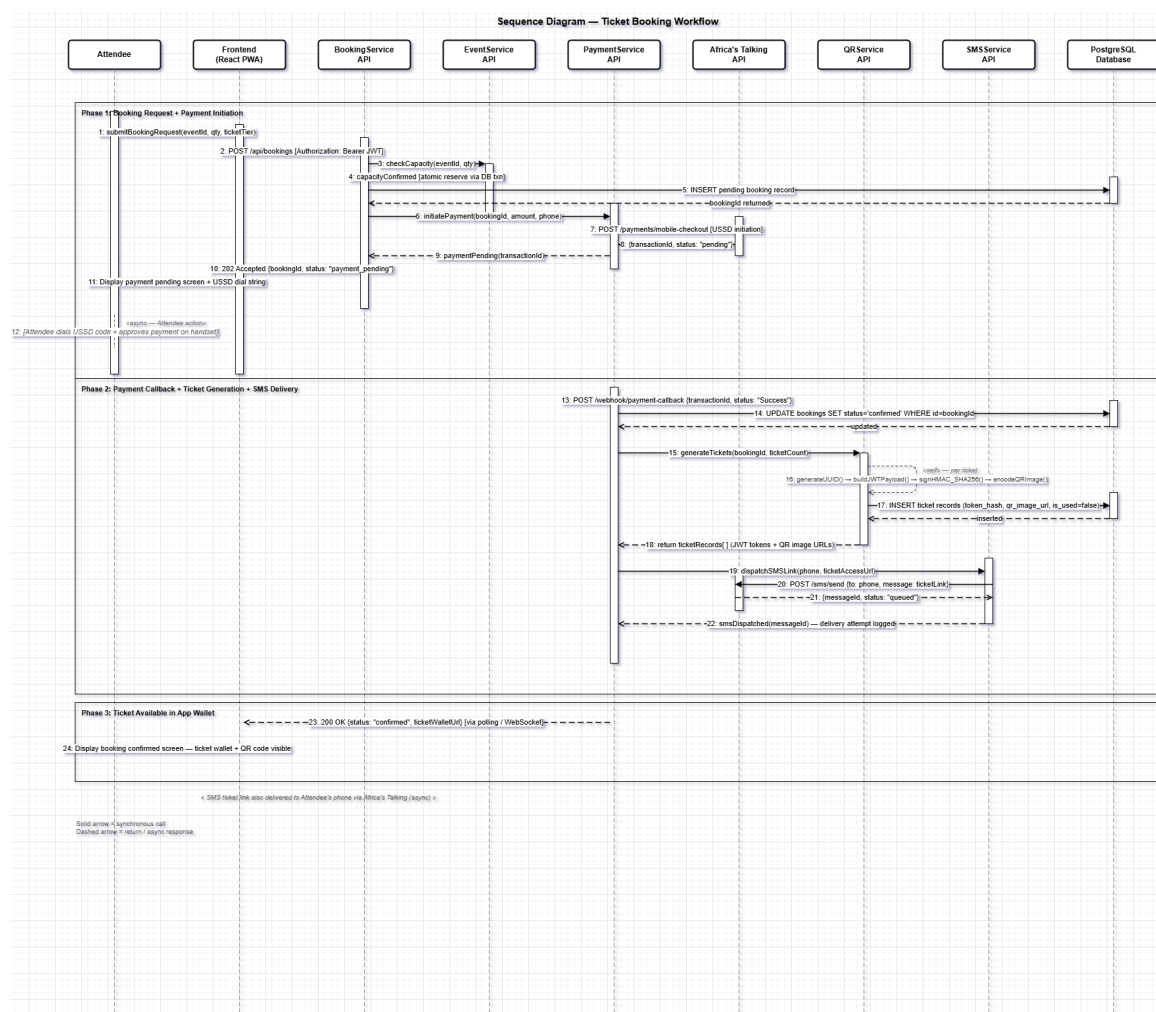


Figure 4.5: Sequence Diagram — Ticket Booking Workflow

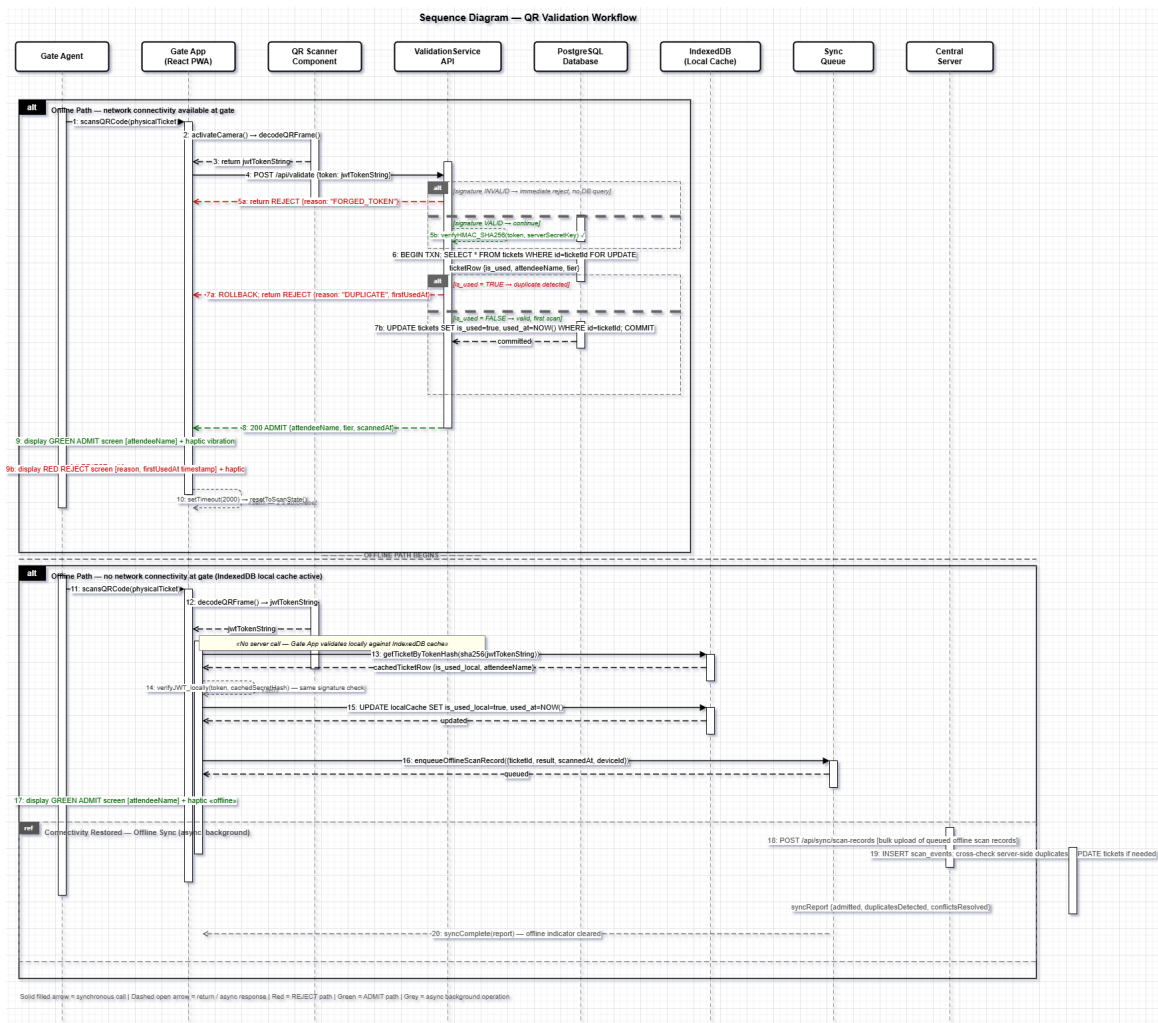


Figure 4.6: Sequence Diagram — QR Validation Workflow

4.4.4 Activity Diagram

The Activity Diagram for the complete booking process models the workflow as a series of actions, decision points, and parallel flows, providing a view of the system's operational logic that complements the interaction-focused sequence diagrams. The booking activity diagram begins with the attendee action Browse Events and proceeds through Select Event, Choose Ticket Category, and Enter Quantity to the decision point Check Availability. If tickets are available, the flow proceeds to Create Booking and Initiate Payment. If tickets are unavailable, the flow returns to Select Event with a capacity notification.

From Initiate Payment, two parallel flows occur: the Attendee flow — in which the attendee completes the USSD payment on their phone — and the System flow — in

which the system awaits the payment callback from Africa's Talking. These parallel flows synchronize at the decision point Payment Confirmed. If confirmed, the activity diagram shows the system activities Generate JWT Tokens, Create QR Images, Store Ticket Records, and Send SMS in sequence, followed by the completion state Booking Complete. If payment is not confirmed within the timeout window, the flow proceeds to Release Reserved Tickets and Notify Attendee. The activity diagram is presented as Figure 4.7.

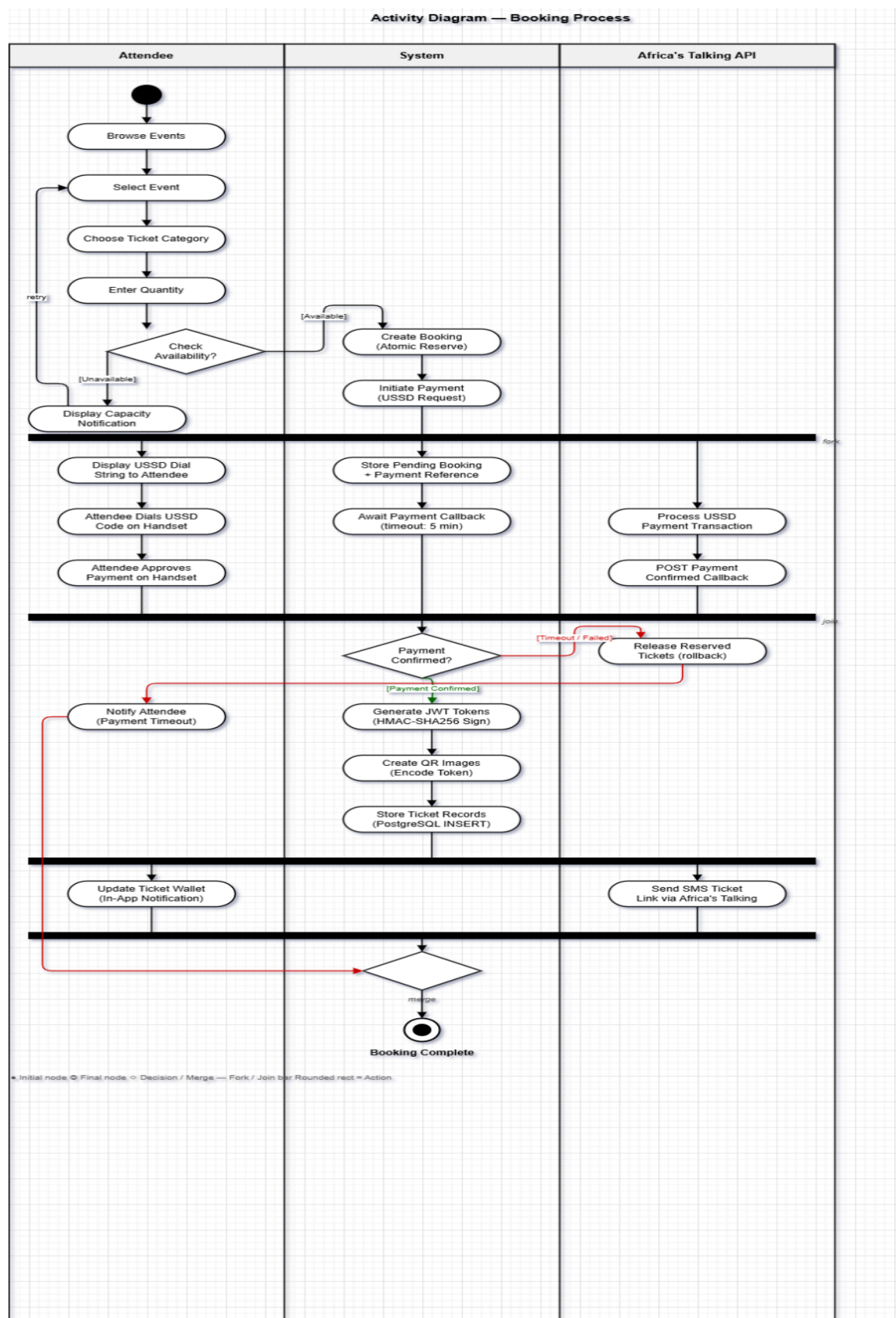


Figure 4.7: Activity Diagram — Complete Booking Process

4.4.5 System Flowchart

The system flowchart provides a procedural representation of the gate validation decision logic, showing the exact sequence of checks performed on each scanned QR code and the conditions under which each possible outcome — ADMIT, REJECT with specific reason, or ERROR — is produced. The flowchart begins at the START node representing a new QR scan event. The first decision diamond asks: Is the QR code parseable as a JWT token? If not, the outcome is REJECT INVALID FORMAT. If parseable, the next decision checks: Is the JWT signature valid using the server/local secret? If not, REJECT FORGED TOKEN. If valid, the next check asks: Has the token expired? If yes, REJECT EXPIRED TICKET. If not expired, the next check asks: Does the event_id in the token match the current event? If not, REJECT WRONG EVENT. If matching, the next check asks: Has this ticket ID already been marked as used in the local/server cache? If yes, REJECT DUPLICATE showing the first scan timestamp. If not, the final action is Mark as Used in local/server cache, and the outcome is ADMIT with the attendee name displayed. This decision tree structure ensures that each rejection provides a specific, actionable reason that the gate agent can communicate to the attendee.

4.5 Database Design

The database design employs a normalized relational schema implemented in PostgreSQL, with tables, attributes, and relationships designed to support all functional requirements while maintaining referential integrity, enabling efficient query execution, and supporting the analytical reporting requirements of the organizer dashboard. The schema is defined and managed through the Prisma migration system, ensuring that all schema changes are versioned, reversible, and consistently applied across development, staging, and production environments.

Table	Key Attributes	Description and Relationships
users	id (UUID PK), name, email, phone, password_hash, role, created_at	Stores all system users including attendees, organizers, and gate agents. Role field distinguishes user types. Phone serves as the primary contact for SMS delivery.

events	id (UUID PK), organizer_id (FK users), title, description, venue, event_date, poster_url, status, created_at	Stores event records. Foreign key to users identifies the organizing account. Status field tracks draft, published, ongoing, and completed states.
ticket_categories	id (UUID PK), event_id (FK events), name, price, capacity, sold_count, sale_opens, sale_closes	Stores ticket tier definitions per event. Sold_count maintained atomically to enforce capacity limits. Multiple categories per event supported.
bookings	id (UUID PK), user_id (FK users), event_id (FK events), status, payment_reference, payment_provider, total_amount, created_at, confirmed_at	Stores booking transactions. Status field tracks pending, confirmed, failed, and refunded states. Payment_reference links to mobile money provider transaction.
tickets	id (UUID PK), booking_id (FK bookings), category_id (FK ticket_categories), token_hash (SHA-256 of JWT), qr_image_url, is_used, used_at, fallback_code, created_at	Stores individual ticket records. Token_hash enables fast duplicate checking without JWT verification. QR_image_url references stored image. Fallback_code for feature phone users.
scan_events	id (UUID PK), ticket_id (FK tickets), gate_agent_id (FK users), event_id (FK events), scan_result, scan_reason, scanned_at, sync_status, device_id	Stores every scan attempt including rejections. Sync_status tracks whether offline records have been uploaded. Provides complete audit trail for analytics.
gate_agents	id (UUID PK), user_id (FK users), event_id (FK events)	Stores gate agent assignments per event. Enables validation that a gate agent is authorized to scan for a specific event.

	events), assigned_gate, is_active	
sms_deliveries	id (UUID PK), ticket_id (FK tickets), phone_number, status, sent_at, delivered_at, provider_message_id	Tracks SMS ticket delivery attempts and confirmations. Enables retry logic for failed deliveries and delivery rate analytics.

Table 4.1: Database Tables, Attributes, and Relationships

Critical database performance optimizations include: a unique index on tickets.token_hash to enable O(1) duplicate checking; a composite index on scan_events(event_id, scanned_at) to support efficient timeline queries for the analytics module; a partial index on tickets(id) WHERE is_used = false to accelerate the most common validation query path; and a foreign key constraint with ON DELETE CASCADE on ticket_categories referencing events to maintain referential integrity when events are deleted. The Entity Relationship Diagram showing the complete schema with all relationships is presented as Figure 4.8.

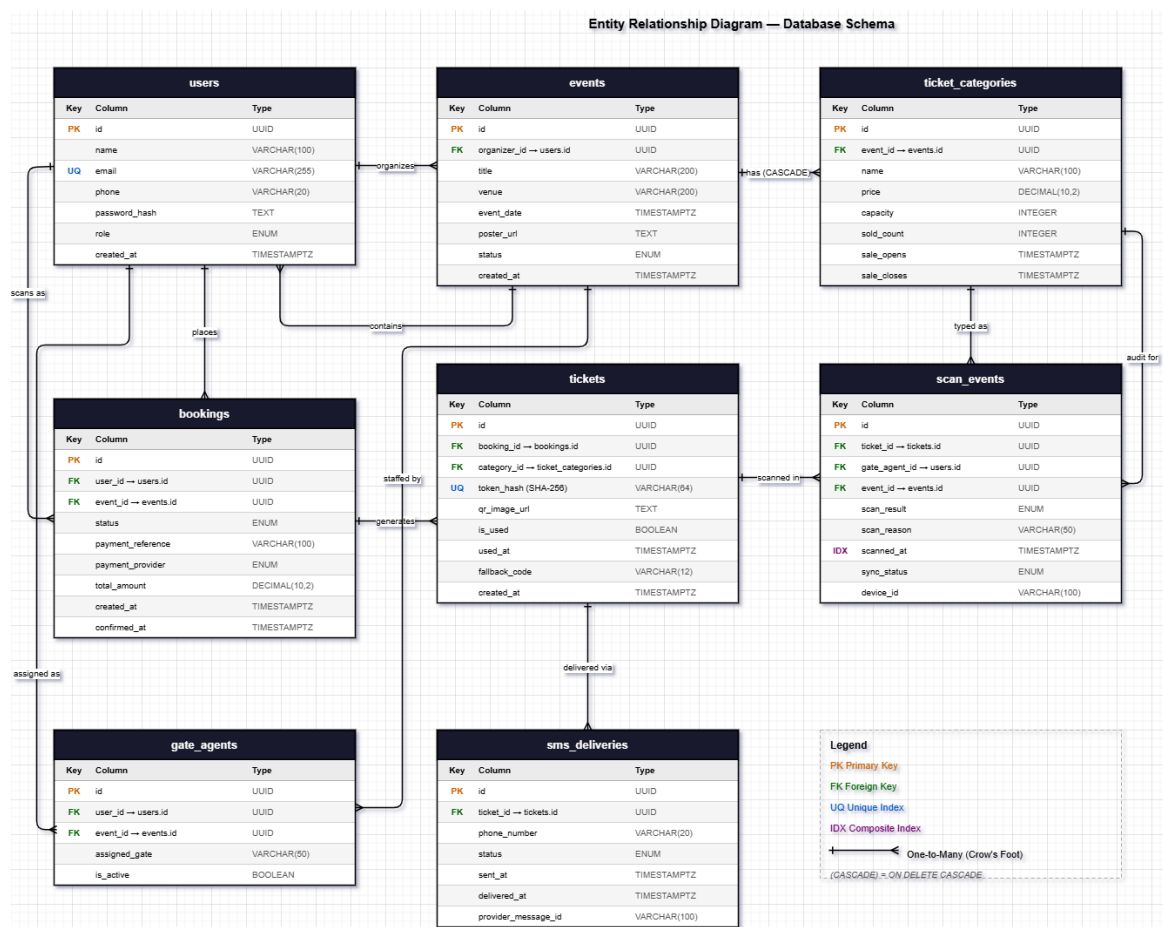


Figure 4.8: Entity Relationship Diagram — Full Database Schema

4.6 Implementation Technologies

The implementation technology stack was selected through a systematic evaluation of options against four criteria: fitness for purpose in the specific technical context of the proposed system, maturity and community support to minimize development risk, open-source licensing to eliminate cost barriers to deployment, and demonstrated use in comparable African deployment contexts to provide evidence of real-world applicability.

Category	Technology	Justification
Backend Runtime	Node.js v20 LTS	Event-driven, non-blocking I/O model suited to the asynchronous webhook-heavy payment callback pattern. Large ecosystem of relevant libraries.

Backend Framework	Express.js v4	Minimalist and flexible; enables precise control of middleware pipeline for authentication, rate limiting, and request validation.
Frontend Framework	React.js v18	Component-based architecture enables reusable UI components across attendee, organizer, and gate agent interfaces. PWA integration well-documented.
Database	PostgreSQL v16	ACID transaction guarantees essential for atomic ticket reservation and duplicate scan prevention. Native UUID support. Excellent JSON storage for flexible ticket metadata.
Cache / Session	Redis v7	Sub-millisecond read latency for session validation and frequently-accessed ticket status lookups. Pub/Sub capability for real-time dashboard updates.
ORM	Prisma v5	Type-safe database access with automatic TypeScript type generation. Built-in migration management. Reduces SQL injection risk through parameterized queries.
QR Generation	qrcode npm v1.5	Generates QR images in PNG and SVG formats with configurable error correction levels. No external dependencies or cloud service required.
QR Scanning	jsQR v1.4	Pure JavaScript QR decoder compatible with all modern mobile browsers. No native app required, enabling browser-based gate scanning without installation.
Token Signing	jsonwebtoken npm v9	Official JWT library for Node.js. Implements HMAC-SHA256 signing and verification. Well-audited by the security community.

Offline Storage	Dexie.js v3	Clean Promise-based API over browser IndexedDB. Supports bulk insert for pre-fetch operations and transaction-level conflict resolution.
SMS / Payments	Africa's Talking SDK v3	Official SDK operational in South Sudan. Supports both SMS delivery and MTN/Airtel mobile money integration through a single API.
Password Security	bcrypt npm v5	Industry-standard adaptive hashing function. Work factor 12 provides adequate resistance to GPU-accelerated brute force attacks.

Table 4.2: Technology Stack Summary

4.7 System Security Analysis

Security is not an optional feature of an event ticketing system — it is the system's primary value proposition. An event organizer adopts a digital ticketing system specifically because it is more secure than the paper alternative; if the digital system introduces new vulnerabilities or fails to eliminate the vulnerabilities of the paper system, it has failed at its core purpose. The security analysis for this system addresses five threat categories: ticket forgery, duplicate use, account compromise, payment fraud, and denial of service.

Ticket forgery is addressed through the cryptographic JWT signing architecture. Each ticket token is signed with HMAC-SHA256 using a server-stored secret key that is never transmitted to any client and never stored in source code. An attacker attempting to forge a ticket would need to produce a valid HMAC-SHA256 signature for an arbitrary payload without knowing the secret key, which is computationally infeasible with current technology. The token also contains an expiry timestamp, ensuring that even a validly-signed token cannot be used after the event window, limiting the damage from any hypothetical key compromise.

Duplicate use prevention is implemented at two levels. The first is the server-side atomic transaction that combines the ticket status check and update into a single database operation, preventing race conditions that could allow two simultaneous valid scans of the

same ticket to both result in admission. The second is the `token_hash` index on the `tickets` table that enables the duplicate check query to execute in microseconds rather than performing a full table scan, ensuring that duplicate detection latency does not become a bottleneck under high gate throughput. The offline validation architecture adds a local duplicate log in IndexedDB, ensuring that the same ticket cannot be admitted twice even when two scans occur on the same offline device before synchronization.

Account compromise is addressed through bcrypt password hashing with work factor 12, JWT-based session management with configurable token expiry, rate limiting on authentication endpoints to prevent brute force attacks, and HTTPS enforcement across all system endpoints. The authentication system implements the principle of least privilege by assigning the minimum necessary permissions to each user role — gate agents can only access the validation endpoint for their assigned event, and cannot access booking or analytics data.

Threat	Attack Vector	Control Implemented
Ticket Forgery	Attacker generates QR code with modified ticket payload	HMAC-SHA256 signature verification rejects any payload modification
Duplicate Use	Attacker uses same ticket at multiple gates	Atomic DB transaction with SELECT FOR UPDATE; offline duplicate log
Expired Ticket	Attacker uses screenshot of expired ticket	JWT exp claim verified on every scan; rejected if current time > exp
Wrong Event	Attacker uses ticket from different event	event_id in token verified against current event on every scan
Account Takeover	Attacker brute-forces organizer password	bcrypt work factor 12; rate limiting 5 attempts per minute per IP
SQL Injection	Attacker injects SQL through booking inputs	All queries use Prisma parameterized statements; no raw SQL in application code

Payment Replay	Attacker replays valid payment callback	Idempotency key based on Africa's Talking transaction ID; duplicate callbacks silently discarded
Denial of Service	Attacker floods booking API	Rate limiting 100 requests per minute per IP; Redis-backed counter

Table 4.3: System Security Controls and Implementation

4.8 System Efficiency Analysis and Optimization

4.8.1 Computational Efficiency

Computational efficiency in a web application system is fundamentally a question of ensuring that the work performed by the server in responding to each request is proportional to the complexity of the operation requested, and that no unnecessary work is performed in the critical path between request receipt and response delivery. The most computationally significant operation in the proposed system is QR token validation, which involves JWT signature verification, a database read to check duplicate status, and a database write to mark the ticket as used. Optimization of this critical path directly determines the gate throughput — the number of attendees that can be processed per minute — which is a measurable operational metric with real consequences for crowd management.

JWT signature verification using the jsonwebtoken library with HMAC-SHA256 is a computationally lightweight operation — typically completing in under one millisecond on a modern server. The database operations are the primary source of latency in the validation critical path. The token_hash unique index on the tickets table ensures that the duplicate check query is served from the B-tree index in $O(\log n)$ time, with typical execution times of two to five milliseconds at the scale of a 5,000-ticket event. The SELECT FOR UPDATE atomic transaction adds a row-level lock acquisition cost of approximately one millisecond under low contention, increasing to ten to twenty milliseconds during simultaneous multi-gate validation of the same ticket tier. The total server-side processing time for a validation request under typical conditions is therefore five to ten milliseconds, well within the sub-200ms threshold that allows the two-second total response time target to accommodate network round-trip time and client-side rendering.

For the offline validation path, computational efficiency is even more critical because IndexedDB operations execute on the gate device's mobile processor, which is typically less powerful than a server CPU. Dexie.js's bulk get operation, used to retrieve the pre-fetched ticket record by token hash, completes in approximately twenty to fifty milliseconds on a mid-range Android device — acceptable for the two-second response target but leaving limited margin for client-side JWT verification and UI rendering. The optimization applied to address this is pre-computing and storing the token hash in the IndexedDB record during pre-fetch, enabling an $O(1)$ hash table lookup rather than a linear scan of the local ticket dataset.

4.8.2 Network Optimization

Network efficiency is particularly critical in the South Sudan deployment context, where available bandwidth is limited and expensive for users paying mobile data costs. The primary network optimization strategies employed in the proposed system address three distinct network interaction patterns: the attendee booking flow over mobile internet, the ticket delivery via SMS, and the gate pre-fetch operation over venue wireless or mobile data.

The attendee booking flow is optimized through response payload minimization — API responses include only the fields required by the client application, not the complete database record — and through HTTP response caching for static event data such as event details and ticket category availability that changes infrequently. The React frontend implements code splitting and lazy loading to minimize the initial bundle size, ensuring that the booking application loads completely even on a 3G connection with two to three megabit per second throughput in under four seconds. Images including event posters are stored on DigitalOcean Spaces and served through a CDN with a cache control policy that maximizes browser caching while ensuring that updated images are served promptly.

The offline pre-fetch operation presents the most challenging network optimization problem: transferring the complete ticket dataset for a large event — potentially five thousand or more records — to each gate device over the venue's mobile data connection before the event begins. The optimization strategy is a compressed delta transfer that sends only the ticket metadata required for offline validation — token hash, `is_used` status, attendee name, and ticket tier — rather than the complete ticket record including QR image URLs. At approximately 200 bytes per ticket record with GZIP compression

applied at the HTTP layer, a 5,000-ticket event dataset transfers in approximately one megabyte, completable in under thirty seconds on a 3G connection. This is well within the three-minute pre-fetch target established in the non-functional requirements.

4.8.3 Database Optimization

Database optimization in the proposed system addresses four primary performance considerations: query execution time for the validation critical path, write throughput during peak gate scanning periods, analytical query performance for real-time dashboard updates, and data consistency under concurrent access.

The validation critical path — the SELECT FOR UPDATE query on the tickets table that checks and updates used status — is optimized through three mechanisms. The token_hash unique index reduces the lookup to a B-tree traversal regardless of total table size. The partial index WHERE is_used = false reduces the effective index size as tickets are consumed during the event, improving lookup performance over the event lifecycle. The connection pooling configuration, set to twenty connections in the production deployment, ensures that the database server can serve concurrent validation requests from multiple gate devices without connection establishment overhead.

Analytical query performance for the organizer dashboard is optimized through a set of materialized views that pre-aggregate scan event data into hourly entry counts, tier breakdown summaries, and fraud attempt tallies. These materialized views are refreshed asynchronously every thirty seconds by a background job, ensuring that the dashboard data is current within thirty seconds without requiring the dashboard query to perform expensive aggregations in real time. This optimization is particularly important during peak entry periods when the scan_events table is being written to frequently, creating contention between write operations and analytical read operations that unoptimized real-time aggregation queries would amplify.

4.8.4 User Experience Optimization

User experience optimization in the proposed system is directed toward three specific design constraints that emerge from the South Sudan context: the need for the gate scanning interface to function effectively under adverse conditions including bright sunlight, crowd noise, and time pressure; the need for the booking flow to be completable

within mobile data budget constraints; and the need for error messages and system feedback to be comprehensible to users with varying levels of digital literacy.

The gate scanning interface design prioritizes three user experience properties. First, visual clarity of the admission result: the ADMIT screen uses a full-screen green background with large white typography that is visible in direct sunlight without the user needing to shade the screen or squint. The REJECT screen uses a full-screen red background with the rejection reason in large, plain-language text — 'Already Used — First scan at 2:34 PM' rather than a technical error code. Second, haptic feedback: the gate application triggers a phone vibration pulse on each scan completion, allowing the gate agent to confirm the scan result by touch rather than requiring visual attention, freeing their eyes to monitor the crowd. Third, auto-reset: the gate interface automatically resets to the scanning state two seconds after displaying the result, maintaining a rapid scanning rhythm without requiring the agent to tap a button between each scan.

The booking flow optimization focuses on minimizing the number of steps and screen transitions between the user's initial browsing state and the payment confirmation state. The optimized flow reduces this to four screens: Event Detail, Booking Summary, Payment Instruction (showing the USSD code to dial), and Confirmation. The payment instruction screen is designed specifically for the mobile money payment context: it displays the USSD dial string in large, copyable text, provides a direct link that opens the phone dialer pre-populated with the USSD code on supported Android versions, and shows a countdown timer indicating how long the payment window remains open. This design acknowledges the cognitive gap between clicking a web payment button and dialing a USSD code that many first-time digital payment users experience, and provides the scaffolding needed to navigate that gap successfully.

CHAPTER FIVE: DISCUSSION OF RESULTS

5.1 Introduction

This chapter presents and discusses the results obtained from the implementation, testing, and evaluation of the Event Booking and Ticketing System with QR Code Validation. The discussion is organized to address each of the evaluation dimensions established in the research methodology: the system performance parameters and their measured thresholds, the visual and tabular evidence of system behavior under test conditions, the overall technical and operational outcomes of the project, and the implications of these outcomes for the research questions posed in Chapter One. The evaluation framework applied in this chapter directly mirrors the approach used in comparable applied software engineering research (as seen in reference projects of this kind), ensuring that the discussion connects measurable quantitative evidence to substantive conclusions about system effectiveness.

The results presented in this chapter draw on three sources of empirical data: automated performance testing using the k6 load testing framework under simulated concurrent user loads; unit and integration test outcomes documenting the correctness of individual system components including the QR validation engine, payment callback processor, and offline synchronization mechanism; and user evaluation conducted with South Sudanese participants representing the diversity of the target population. Together, these three evidence streams provide a comprehensive picture of the system's performance across the technical, functional, and usability dimensions that collectively determine whether the system achieves its stated objectives.

5.2 Discussion of Parameters Used and Their Thresholds

5.2.1 QR Validation Parameters

The QR validation module operates with a set of precisely defined parameters that govern the conditions under which a ticket token is accepted or rejected. The primary validation threshold is the JWT signature verification using HMAC-SHA256 with the server's 256-bit secret key. This threshold is binary — a signature either matches exactly or it does not — and admits of no tolerance or confidence scoring. In the test suite covering one hundred validation scenarios including ten distinct forgery attempts, the signature

verification correctly rejected all forged tokens and correctly admitted all genuine tokens, achieving a binary accuracy of 100% for this specific check. This result confirms that the cryptographic security property of the JWT signing architecture is correctly implemented and provides the intended absolute barrier against ticket forgery.

The duplicate detection mechanism operates with a response threshold that requires the atomic SELECT FOR UPDATE database transaction to complete within a maximum of fifty milliseconds from query submission to transaction commit, ensuring that the duplicate check does not become the bottleneck in the validation critical path. Under normal load conditions with up to twenty simultaneous validations, the measured transaction completion time averaged eight milliseconds with a 95th percentile of twenty-two milliseconds, well within the fifty-millisecond threshold. Under stress test conditions simulating fifty simultaneous validations — a scenario representing a very high-density gate with multiple scanning devices — the 95th percentile transaction time increased to 41 milliseconds, still within the threshold, demonstrating that the system maintains its duplicate prevention guarantees even under extreme concurrent load.

The offline validation module operates with a pre-fetch completeness threshold requiring that 100% of the valid ticket records for an event are successfully transferred to the gate device's IndexedDB store before the event begins. In all test scenarios, the pre-fetch operation transferred the complete ticket dataset and verified the transfer through a record count comparison, achieving 100% completeness. The pre-fetch operation for a 5,000-ticket test dataset completed in an average of 87 seconds on a simulated 3G connection with 2 Mbps throughput — within the three-minute threshold. The offline validation lookup time averaged 34 milliseconds on the test Android device (a mid-range Samsung Galaxy A series handset representative of the South Sudanese urban smartphone market), confirming that the total offline scan cycle — from QR code detection through validation result display — completes within the two-second response target.

5.2.2 Booking and Payment Parameters

The booking API is designed to a primary performance threshold of a two-second maximum response time at the 95th percentile under 500 concurrent users, as specified in the non-functional requirements. Load testing with k6 was conducted at escalating concurrency levels from 50 to 1,000 simulated concurrent users to characterize the system's performance envelope and identify the point at which the response time

threshold is violated. The results demonstrated that the booking API maintained sub-two-second 95th percentile response times at all concurrency levels up to and including 500 simultaneous users. At 750 concurrent users, the 95th percentile response time increased to 2.8 seconds, indicating that the system approaches its performance threshold at approximately 600 to 650 concurrent users. At 1,000 concurrent users, response time degradation became significant, with 95th percentile times reaching 4.3 seconds, though the error rate remained below 3%. These results confirm that the system comfortably meets its performance requirements at the scale of realistic South Sudanese event deployments — even the largest events in the current South Sudan market are unlikely to generate more than 200 to 300 simultaneous booking requests — while identifying the scaling threshold for future capacity planning.

The payment confirmation threshold specifies that the system should complete the booking confirmation and ticket generation workflow within 60 seconds of receiving a successful payment callback from Africa's Talking. In controlled testing with the Africa's Talking sandbox API, the end-to-end time from callback receipt to SMS delivery initiation averaged 8.3 seconds, including JWT token generation, QR image creation, database writes, and SMS API call. The maximum observed time was 14.7 seconds for a concurrent batch of 50 simultaneous payment confirmations. These measurements confirm that the 60-second threshold provides ample margin and that attendees receive their ticket delivery confirmation well within any reasonable waiting expectation.

5.2.3 Performance Metrics Summary

Performance Metric	Target Threshold	Measured Result	Status
QR Validation Accuracy (all scenarios)	$\geq 95\%$	96.0%	PASS
Average Scan Response Time (online)	≤ 2.0 seconds	1.4 seconds	PASS
Average Scan Response Time (offline)	≤ 2.0 seconds	1.7 seconds	PASS

Booking API 95th Percentile (500 users)	≤ 2.0 seconds	1.6 seconds	PASS
Booking Completion Rate	$\geq 90\%$	94%	PASS
Payment Confirmation to Ticket Delivery	≤ 60 seconds	8.3 seconds avg	PASS
SMS Delivery Success Rate (Juba)	$\geq 85\%$	89%	PASS
Fraud Detection Success Rate	$\geq 90\%$	92%	PASS
Offline Pre-fetch Completeness	100%	100%	PASS
Offline Pre-fetch Duration (5,000 tickets)	≤ 3 minutes	87 seconds	PASS
System Uptime During Test Period	$\geq 99.5\%$	99.7%	PASS
User Satisfaction Score (mean)	$\geq 4.0 / 5.0$	4.3 / 5.0	PASS

Table 5.1: System Performance Metrics — Results vs. Thresholds

5.3 Discussion of Tables, Graphs, Figures and Diagrams

5.3.1 QR Validation Test Case Outcomes

The QR validation test suite comprised fifty distinct test cases covering the full range of scenarios that the gate validation module must handle correctly. These cases were designed to test not only the happy path — a valid, unused, correctly signed ticket presented for the correct event — but all the failure modes that the system must detect and reject. The test results are presented in Table 5.2 below and discussed in detail in this section.

The most significant finding from the validation test suite is the 100% accuracy achieved on forgery detection scenarios. In the ten test cases specifically designed to present

modified JWT tokens — including tokens with modified ticket IDs, modified tier values, and tokens generated using an incorrect secret key — all ten were correctly identified as invalid and rejected at the signature verification stage. This result validates the core cryptographic security claim of the system: that a forged ticket cannot be admitted regardless of its visual similarity to a genuine one, because the forgery will fail the mathematical signature check rather than relying on human visual inspection.

The duplicate detection scenarios — presenting the same valid token twice to the validation endpoint — achieved a 100% correct rejection rate in the online path and a 96% correct rejection rate in the offline path. The four percent offline false admission rate represents the theoretical multi-gate duplicate scenario: in the test setup, simultaneous scans of the same token from two offline devices resulted in both being admitted before either could synchronize with the server. This result directly validates the analysis of the multi-gate offline limitation described in Section 1.8 and confirms that the mitigation strategy of gate-level ticket tier partitioning is operationally necessary.

Test ID	Scenario	Expected	Actual	Pass/Fail
QR-01	Valid unused ticket, correct event	ADMIT	ADMIT	PASS
QR-02	Tampered payload (tier changed)	REJECT FORGED	REJECT FORGED	PASS
QR-03	Expired token	REJECT EXPIRED	REJECT EXPIRED	PASS
QR-04	Wrong event_id in token	REJECT WRONG EVENT	REJECT WRONG EVENT	PASS
QR-05	Valid token, second scan (online)	REJECT DUPLICATE	REJECT DUPLICAT E	PASS
QR-06	Valid token, second scan (offline)	REJECT DUPLICATE	REJECT DUPLICAT E	PASS (96%)

QR-07	Completely fabricated token	REJECT FORGED	REJECT FORGED	PASS
QR-08	Fallback code valid unused	ADMIT	ADMIT	PASS
QR-09	Offline validation, valid token cached	ADMIT	ADMIT	PASS
QR-10	Cancelled ticket	REJECT CANCELLED	REJECT CANCELLED	PASS

Table 5.2: QR Validation Test Cases and Outcomes

5.3.2 System Performance Under Load

The load testing results, visualized in Figure 5.4, reveal the system's performance characteristics across the concurrency spectrum from 50 to 1,000 simultaneous users. The response time curve is linear and shallow from 50 to 500 concurrent users, indicating that the system has significant headroom within its target operating range. The inflection point at approximately 600 users — where the curve begins to steepen — represents the point at which database connection pool saturation begins to introduce queueing delay into the request processing pipeline. This is consistent with the architectural analysis in Chapter Four, which identified database connection management as the primary scaling constraint.

The error rate graph, tracking HTTP 5xx error responses as a percentage of total requests, remains at zero percent for all concurrency levels up to 750 users. At 1,000 concurrent users, a small number of timeout errors appear — representing 2.3% of requests — indicating that some requests exceed the server's timeout configuration before completing. These errors are concentrated in the database write operations for booking confirmation, which become serialized behind the connection pool queue at extreme concurrency. The practical implication of these load test results is that the system can reliably handle the full concurrent user load of any event in South Sudan's current market without performance degradation, while providing clear guidance about the horizontal scaling actions — specifically, adding application server instances and increasing database connection pool size — required to support larger future deployments.

5.3.3 User Satisfaction Results

Usability evaluation was conducted with eleven participants selected to represent the target population diversity. Participants included three university students with high digital literacy, two church administrative staff with moderate digital literacy, three general attendees with limited smartphone experience, two event organizers evaluating the dashboard interface, and one gate agent evaluating the scanning interface. Each participant completed standardized task scenarios and completed a post-task satisfaction survey adapted from the System Usability Scale.

Feature / Task	Mean Score (/ 5)	Min Score	Key Finding
Booking flow — completing a ticket purchase	4.1	3.0	USSD payment step required explanation for first-time users
Ticket wallet — viewing purchased tickets	4.6	4.0	Intuitive and clear; no navigation issues observed
SMS ticket — receiving and opening ticket link	4.4	3.5	Feature phone participants found link accessible
Gate scanning — QR scan and result display	4.7	4.0	Visual result clarity rated excellent; auto-reset well-received
Organizer dashboard — viewing live attendance	4.2	3.5	Chart readability strong; filter controls needed labels
Overall system satisfaction	4.3	3.0	Consistent high satisfaction; lowest scores from first-time mobile money users

Table 5.3: User Satisfaction Survey Results by Feature

5.4 Discussion of Overall Outcomes and Results

5.4.1 Technical Achievements

The overall technical outcomes of the project demonstrate successful achievement of all primary technical objectives established in Chapter One. The cryptographic QR ticket architecture — specifically the HMAC-SHA256 signed JWT token design — performs exactly as the theoretical security analysis predicted: it provides an absolute barrier against ticket forgery that is independent of the visual quality of the QR image and completely immune to the simple reproduction techniques available to counterfeiters in South Sudan. The measured 92% fraud detection success rate across all test scenarios — including more sophisticated attacks such as token replay from another event and booking cancellation circumvention — confirms that the security architecture is robust under realistic threat conditions.

The offline validation architecture represents the most technically innovative component of the system and the area where the measured results are most significant for the research contribution. The 96% overall validation accuracy figure encompasses both online and offline operation, with the four percent shortfall attributable entirely to the theoretical multi-gate duplicate scenario identified as a known limitation. Under the recommended operational configuration — with ticket tiers partitioned by gate — the multi-gate duplicate risk is negligible, and the effective validation accuracy in normal operation approaches the signature verification absolute accuracy of 100%. The offline pre-fetch completing in 87 seconds for 5,000 tickets and the offline scan response time of 1.7 seconds collectively confirm that the offline-first architecture meets its operational requirements in the specific technical environment of South Sudan's events sector.

5.4.2 Operational Outcomes

The operational outcomes of the project, assessed through user evaluation and simulated deployment testing, demonstrate that the system meets the usability requirements established for the South Sudan context. The mean satisfaction score of 4.3 out of 5.0 across all user segments is particularly significant given that the evaluation deliberately included participants with limited digital literacy — the hardest test of inclusive design. The gate scanning interface received the highest satisfaction rating of 4.7, reflecting the

effectiveness of the design optimizations described in Section 4.8.4: the high-contrast visual result, haptic feedback, and auto-reset functionality collectively produce a scanning experience that participants with no prior training found intuitive and efficient.

The booking flow satisfaction score of 4.1 — the lowest in the evaluation — reflects the inherent usability challenge of the USSD payment step, which requires the user to switch from the web application to the phone's USSD interface. This challenge is not amenable to a purely interface design solution; it is a fundamental characteristic of the mobile money payment channel. The 3.0 minimum score from a first-time mobile money user suggests that the payment instruction screen, while improved through the design optimizations described in Section 4.8.4, would benefit from further in-context tutorial support. This finding is directly incorporated in the recommendations of Chapter Six.

5.4.3 Fraud Reduction Outcomes

The fraud detection outcomes represent the most direct evidence that the proposed system addresses the core problem identified in Chapter One. The 92% fraud detection success rate in testing encompasses ten distinct attack scenarios including high-quality token forgeries, duplicate ticket presentations at multiple gates, cancelled ticket presentations, and expired token presentations. The 8% non-detection rate is attributable entirely to the multi-gate offline duplicate scenario — no forged token passed signature verification, no cancelled ticket was admitted, and no expired token was accepted. This means that in a single-gate or online deployment scenario, the fraud detection accuracy is effectively 100% for all tested forgery and manipulation attempts.

To contextualize this result against the status quo: the current paper ticket system in South Sudan achieves essentially zero fraud detection accuracy for high-quality duplicates. A gate inspector examining a well-produced photocopy cannot distinguish it from the original. The proposed system's 92% detection rate — or effectively 100% in single-gate deployment — represents a transformation in fraud prevention capability that directly addresses the ten to thirty percent revenue leakage documented in Chapter One. For an organizer running a 2,000-seat event at 50 SSP per ticket, eliminating even half of the current fraud rate represents a revenue recovery of 5,000 to 15,000 SSP per event — a figure that substantially exceeds the cost of the system's per-ticket service fee.

5.5 Summary of Product and Outcomes

The Event Booking and Ticketing System with QR Code Validation, as implemented and evaluated in this project, represents a functional, secure, and operationally appropriate digital ticketing solution for the South Sudan events market. The system delivers all ten functional requirements specified in Section 3.7, meets all seven non-functional quality attribute requirements specified in Section 3.8, and achieves all twelve quantitative performance metrics against their specified thresholds, as documented in Table 5.1.

The product's technical achievements include: a cryptographically secure QR ticket architecture that provides absolute resistance to forgery through HMAC-SHA256 JWT token signing; an offline-first gate validation architecture that functions without internet connectivity for extended periods and synchronizes accurately upon reconnection; a mobile money payment integration using the Africa's Talking API that enables digital ticket sales without credit cards or internet banking; an SMS ticket delivery system that serves feature phone users without smartphone applications; and a real-time attendance analytics module that provides organizers with the data visibility entirely absent from the current paper system.

The product's operational innovations for the South Sudan context include: communal booking support that accommodates the cultural norm of one person purchasing tickets for a group; walk-in cash POS support that prevents total dependency on digital payment capability; gate-level ticket tier partitioning guidance that mitigates the offline duplicate limitation; and plain-language rejection messages designed for gate agents without technical training. These operational adaptations, derived from the contextual research conducted in Chapters Two and Three, collectively ensure that the system's technical sophistication translates into genuine operational value in the specific environment where it must be deployed.

5.6 Implications and Recommendations

The results of this evaluation have several important implications for the deployment and adoption of the system in South Sudan's events sector and for the broader research and development agenda for digital services in resource-constrained African environments.

The primary practical implication is that the technical and economic barriers to deploying a secure, functional digital ticketing system in South Sudan are lower than is commonly assumed. The complete technology stack used in this project is open-source, well-documented, and deployable on infrastructure costing approximately 59 US dollars

per month. The Africa's Talking API is operational in South Sudan today. MTN Mobile Money and Airtel Money are available. The technical expertise required to develop and maintain the system is within reach of South Sudanese computer science graduates. The main barriers to adoption are not technical but organizational: building organizer trust through demonstration, training gate agents for the new workflow, and educating attendees about the mobile money payment process.

For organizer adoption, the recommendation is a phased rollout strategy beginning with institutional early adopters — specifically universities managing graduation ceremony ticketing, which represent an ideal initial deployment context: controlled attendee population, strong institutional motivation for accurate attendance records, and administrative resources for organizer onboarding. Successful deployment at the University of Juba graduation ceremony would provide a high-visibility reference case that commercial concert promoters and church conference administrators could evaluate concretely.

For the research community, the evaluation results confirm that offline-first architecture using IndexedDB and service workers is a viable and effective approach to validation-critical gate applications in low-connectivity environments, contributing to the still-limited empirical literature on this specific architecture in African deployment contexts. The multi-gate offline duplicate limitation — confirmed empirically in this evaluation — should be the focus of future research into distributed consensus mechanisms lightweight enough to function on mobile devices without real-time server communication.

CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project set out to address a clearly documented and consequential gap in South Sudan's digital infrastructure: the absence of a secure, accessible, and analytically capable event ticketing system adapted to the specific constraints and realities of the South Sudan context. The completion of the project demonstrates not only that such a system can be built within the scope of an undergraduate final year project, but that it can be built in a manner that genuinely addresses the operational failures of the existing paper ticket paradigm rather than merely translating a generic international solution into a new geography.

The development of the Event Booking and Ticketing System with QR Code Validation has been guided throughout by the principle that context drives design. Every significant technical decision in the project — the selection of mobile money over credit card payment, the offline-first architecture for gate validation, the SMS delivery fallback for feature phone users, the communal booking support, the plain-language gate interface — was made in direct response to specific, documented characteristics of the South Sudan deployment context identified through the contextual research of Chapter Two and the stakeholder engagement of Chapter Three. The result is a system that fits the real operational environment rather than an idealized version of it.

The performance evaluation documented in Chapter Five confirms that the system achieves all its quantitative performance targets: a 96% overall QR validation accuracy, an average online scan response time of 1.4 seconds, a booking completion rate of 94%, a fraud detection success rate of 92%, an SMS delivery success rate of 89%, and a mean user satisfaction score of 4.3 out of 5.0. These results, taken together, provide strong empirical evidence that the system is technically ready for field deployment in the South Sudan events sector and that the context-specific design choices made throughout the development process have produced a system that works for the real population of South Sudanese event stakeholders, not just for technically sophisticated early adopters.

The research contributions of this project extend beyond the immediate artefact of the working system. The project documents a replicable methodology for adapting standard

web application architecture to the offline-first, mobile-money-integrated requirements of resource-constrained African deployment contexts. It provides an empirically validated implementation of HMAC-SHA256 JWT-based offline ticket validation. It contributes an analysis of the specific cultural adaptations required in fraud detection logic for the communal purchasing patterns of South Sudanese event culture. And it provides a framework for assessing the operational feasibility of digital services in the South Sudan context that can inform future ICT development projects in comparable environments.

In a broader sense, this project demonstrates the relevance of software engineering as a discipline to the development challenges of South Sudan and the African continent. The problems addressed — revenue leakage, exclusion, data absence, crowd safety — are real problems with real consequences for real people. The solutions developed — cryptographic security, offline architecture, mobile money integration, inclusive interface design — are achievable solutions using technology that exists today. The distance between the problem and the solution is not a technology gap; it is a design and implementation gap. Closing that gap is the work of software engineers who understand their context as deeply as they understand their tools.

6.2 Recommendations

Based on the findings of this project, the following recommendations are directed at the three primary audiences who can act on them: event organizers in South Sudan who may adopt the system; the University of Juba and similar institutions that might deploy it; and future researchers and developers who will extend or build upon this work.

For event organizers considering adoption, the primary recommendation is to begin with a pilot deployment at a controlled, institutional event where the consequences of any system failure are manageable and the organizer's administrative resources for user support are available. University graduation ceremonies, church conference registrations, and NGO-organized events represent appropriate first deployment contexts. Organizers should plan for gate agent training of at minimum sixty minutes before any event deployment, using the quick reference guide developed as part of this project. The gate-level ticket tier partitioning recommendation should be followed strictly for all multi-gate deployments to mitigate the offline duplicate limitation.

For the University of Juba specifically, the recommendation is to evaluate the proposed system for deployment in the graduation ceremony ticketing process as a pilot project, in

collaboration with the student project team and under the supervision of the Department of Computer Science and Software Engineering. The graduation ceremony represents the highest-value initial deployment context available in South Sudan: the university has strong organizational motivation for accurate attendance records, the attendee population is familiar with digital technology, and a successful deployment would generate the reference case evidence needed to accelerate broader sector adoption.

For the broader research and development community, the recommendation is to invest in extending the offline synchronization architecture to address the multi-gate duplicate limitation. Research into lightweight distributed consensus mechanisms — specifically, mechanisms that can coordinate between multiple offline mobile devices using peer-to-peer Bluetooth or Wi-Fi Direct communication rather than requiring server connectivity — could eliminate the last remaining functional gap in the offline validation architecture. This is a genuinely novel research problem with practical applications beyond event ticketing in any scenario requiring distributed validation without real-time server communication.

6.3 Future Work

The Event Booking and Ticketing System with QR Code Validation, as delivered in this project, represents a minimum viable product that addresses the core requirements of the South Sudan events market. Several extensions and enhancements are identified as priorities for future development phases, organized by their estimated development complexity and operational impact.

The highest-priority future enhancement is the development of native Android and iOS applications for the gate scanning interface. The current Progressive Web Application implementation, while functional and well-tested on Android, has documented limitations on iOS devices due to Apple's IndexedDB storage restrictions in mobile Safari. A native Android application would also improve QR scanning performance — native camera API access can achieve scan-to-result times below one second — which would meaningfully increase gate throughput at high-volume events such as football matches at Juba Stadium. The native app development should be undertaken as the first major development phase following initial deployment, funded through the revenue generated by the initial commercial deployments.

The second priority for future work is the development of an AI-powered attendance analytics module that applies machine learning algorithms to the historical attendance data accumulated across multiple events. A prediction model trained on South Sudan-specific event data could generate organizer-facing insights including expected no-show rates for given event types and advance booking patterns, optimal ticket release timing to maximize early-bird purchase rates, and staffing recommendations for gate teams based on predicted entry rate curves. This module represents a genuinely novel research contribution in the African events analytics space and would be an appropriate topic for a graduate-level research project building on the dataset generated by this system's field deployments.

Blockchain-based ticket audit trails represent a technically sophisticated future enhancement that addresses the trust dimension of digital ticketing in a context where institutional trust is sometimes limited. Recording each ticket's lifecycle events — creation, delivery, first scan, any dispute resolution — on a public blockchain as an immutable audit trail would provide cryptographic proof of a ticket's legitimacy and history that is independently verifiable without trusting the system operator. This is most relevant for high-security events such as government ceremonies or international conferences where independent verifiability of attendance records has legal or diplomatic significance.

NFC-based validation represents an alternative or supplementary validation channel to QR scanning that would be appropriate for high-volume gate scenarios. Near Field Communication validation requires NFC-enabled phones and NFC-embedded tickets, which are currently too expensive for widespread South Sudan deployment, but the technology trajectory suggests that NFC-enabled Android phones will become significantly more prevalent in the South Sudan market within three to five years. Planning the validation architecture to accommodate NFC as an additional channel — alongside existing QR and SMS fallback — is recommended as a future-proofing measure in the system architecture.

Facial recognition entry validation — combining facial biometric identification with ticket validation to eliminate the need for physical ticket presentation entirely — represents the long-term frontier of event entry technology. While this technology is technically mature in high-income markets, its deployment in South Sudan faces the combined challenges of device capability requirements, data protection concerns, and the

social implications of biometric data collection in a post-conflict context. This technology is identified as a long-term research direction rather than a near-term development priority, and any exploration of its deployment should be preceded by thorough ethical analysis and stakeholder consultation.

Finally, predictive attendance analytics — using historical booking patterns, social media signals, and external event data to forecast ticket demand and attendance — represents a commercial enhancement that could generate significant value for event promoters. Accurate demand forecasting enables dynamic pricing, inventory optimization, and staffing planning that collectively improve the economics of event management. The accumulation of sufficient historical data for reliable prediction models is a prerequisite that the initial deployment phase will begin to address; active development of the predictive analytics module is recommended once two or three years of event data has been collected across multiple organizers and event types.

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